Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

The format of spatial memory in the hippocampus varies systematically with a species' visual ecology, but whether this reflects a general sensor–memory coupling is hard to test directly: animal sensor ecology is fixed within species, and AI systems rarely co-train encoder and memory along a sensor axis. We test this in silicon: a navigation agent where only the visual sensor varies along an axis of encoder spatial bandwidth, with architecture, task, and training held fixed. Across five conditions, agents reach near-identical task performance yet the hidden-state code for position differs in form. With a vision-limited encoder, position concentrates into a scene-invariant direction linearly readable from the hidden state; with a rich encoder, it distributes into non-linear directions and partly offloads to the encoder. Linear readability shifts faster than total position information — a format shift, not a total-information collapse. We frame this as an emergent *capacity allocation* between the encoder route and the memory-integration route, measured along three orthogonal axes (magnitude, format, consumption). Memory transplants are asymmetric — rich-encoder policies fail on bottleneck states, the reverse works — paralleling cross-modality remapping in hippocampal coding under hidden-state inference. Probe-readable position further dissociates from policy reliance: a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute. This positions sensor structure as a training-time lever on cognitive-map format, generates a falsifiable architecture-agnostic prediction for transformer navigators, and reframes bandwidth-restricting perception as potentially cognition-enabling rather than only compute-saving. Beyond this empirical result, we treat our matched five-condition silicon analogue as a controlled testbed for porting the systems-neuroscience analytic toolkit — population geometry, intrinsic timescales, fixed-point dynamics, predictive-coding residuals, splitter-cell mixed selectivity, persistent homology of the state manifold — to artificial navigation agents, providing the controlled counterpart to cross-species hippocampal comparisons that biological neuroscience cannot easily achieve.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a pipeline: scale the encoder, and any downstream module benefits. Recent benchmarks complicate this picture — vision–language models with state-of-the-art surface visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025], with two competing diagnoses: a training-distribution gap [Chen et al., 2024], and an encoder-side argument that pretrained encoders carry markedly different geometric content depending on training objective [Tong et al., 2024, El Banani et al., 2024]. We pursue an even sharper structural reading: the encoder and the memory downstream of it are a co-trained system, and *the encoder's structure shapes what is linearly readable from the memory about space* — a content-level claim about the perception–memory interface.

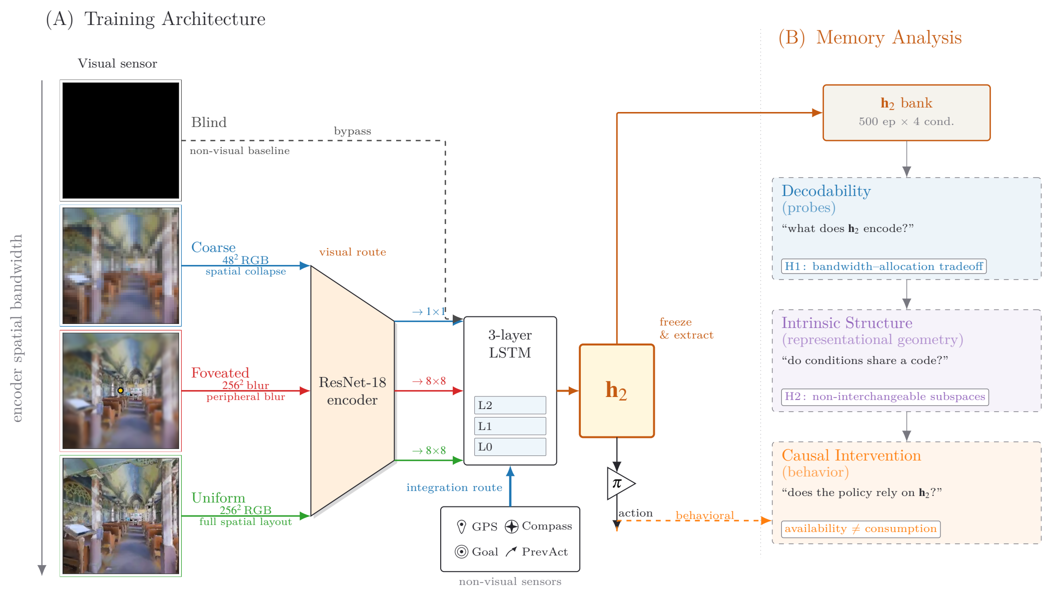


Figure 1: **Experimental setup and analysis pipeline.** *Row 1 (architecture):* four sensor conditions span an axis of encoder spatial bandwidth (*blind*, *coarse*, *foveated*, *uniform*; leftmost column), feeding a shared ResNet-18 encoder + 3-layer LSTM; *blind* bypasses the encoder. A log-polar foveation control (Appendix G) adds a 5th intermediate point on the same axis. *Row 2 (analysis):* three method blocks — decodability, intrinsic structure, causal intervention — read the frozen top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ to test three axes of one capacity allocation: *magnitude*, *format*, *consumption* (§2; synthesis in §4).

Cognitive neuroscience offers concrete precedent for this at the species level: primate view-cells under foveated saccades versus rodent place-cells under near-panoramic vision [Wirth et al., 2017, Rolls, 2024]; cross-modality remapping in echolocating bats whose hippocampal coding reorganises between vision and echolocation [Geva-Sagiv et al., 2016]; congenitally blind humans showing larger anterior hippocampal volume and supranormal non-visual navigation [Fortin et al., 2008]. Whether this generalises to artificial systems is hard to test in animals — their sensor ecology is fixed within species, and modern AI systems rarely co-train encoder and memory from scratch along a sensor axis. We test it in silicon, where this design degree of freedom is available.

Beyond the specific bandwidth–allocation hypothesis, the matched-condition design we adopt opens a methodological avenue that biological neuroscience cannot easily realise. Comparative hippocampal studies (rats, bats, monkeys, blind humans) cannot hold sensor structure independent of evolutionary lineage, body plan, or learning history; cross-species effects are confounded with everything else that differs across species. In a silicon analogue this confound dissolves: encoder, recurrent module, action repertoire, training task, optimiser, and reward structure are identical, and only the sensor varies. This invites the systems-neuroscience analytic toolkit — population probes, intrinsic timescales, dynamical-systems portraits, mixed-selectivity / splitter analyses, predictive-coding residuals, persistent-homology of the state manifold — to be applied to a *controlled comparative population*, not to a single recording. Our paper treats this comparative methodology as an explicit secondary contribution: alongside the empirical capacity-allocation result, we report a set of cogneuro-style analyses on the five-condition population, framing the work as a step toward a comparative cognitive neuroscience of artificial navigation agents.

Cognitive-map-like representations emerge in the recurrent state of deep-RL navigation agents even with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018],[1] but the literature studies one condition at a time, leaving open how encoder structure interacts with what the

[1]"Cognitive map" in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept [Epstein et al., 2017] are engaged by our LOSO scene-invariance test (relational across rooms) and predictive-horizon analysis (SR-style future-state code, Stachenfeld et al. 2017); hierarchical scene clustering and topological reachability are deferred to follow-up.

memory builds. We adopt this agent family as a comparative chassis (Figure 1): identical encoder–memory–policy pipeline, task, and training; varying only the visual sensor across five conditions on an axis of encoder spatial bandwidth (§2).

If the visual encoder and the recurrent integration of GPS-sensor history are competing routes for a finite recurrent-memory budget, *then* richer encoders should erode the linear-readable position signal in the recurrent state while leaving task performance intact. We observe exactly this: across the five sensor conditions, top-layer linear GPS R^2 falls monotonically with encoder spatial bandwidth while success rate stays at 93–99%. We frame this as an emergent *capacity allocation* between visual and integration routes and test it along three orthogonal axes: *magnitude* (how much linear position information lives in the recurrent state), *format* (which subspace carries it across conditions), and *consumption* (which route the policy actually uses for behaviour). Cross-method convergence across linear/MLP/MINE probes, leave-one-scene-out validation, Procrustes shape distance, 5×5 memory transplant, and shortcut-discovery tests rules out single-method artefacts; the asymmetric direction-of-transplant result establishes the format claim outside any probe framework [Orozco et al., 2025].

Contributions. (i) A 5-condition controlled silicon model isolating sensor-induced spatial-format change while holding encoder backbone, task, training, and reward fixed. (ii) The *capacity-allocation* principle: a single mechanism organising five concomitant signatures (magnitude collapse, format divergence, consumption dissociation, scene-invariance, place-vs-view-cell character) into three orthogonal axes. (iii) An asymmetric memory-transplant test (rich-encoder policies fail to use bottleneck states; bottleneck policies still drive themselves from rich-encoder ones) that mirrors hippocampal cross-modality remapping at the behavioural level. (iv) A falsifiable architecture-agnostic prediction for transformer-based navigators (§4, cf. Whittington et al. 2022). For visual intelligence, capacity allocation offers an architecture-side complement to data-side accounts [Chen et al., 2024] of VLM spatial-reasoning failures, and reframes bandwidth-restricting perception (foveation, log-polar) as potentially cognition-*enabling* rather than only compute-saving.

2 Methods

Architecture, task, and conditions (Figure 1, Row 1). PointGoal navigation in Habitat [Savva et al., 2019] trained with DD-PPO [Wijmans et al., 2020] across five conditions varying only the visual sensor: *blind* (no encoder); *coarse* (48×48 RGB, 1×1 encoder feature map after ResNet-18 downsampling); *foveated* (256×256 with eccentricity-dependent Gaussian blur $\sigma_{\max}$=8, 4×4 feature map); *uniform* (256×256 unblurred, 4×4); *foveated-logpolar* ($\sim 2\times2$ via log-polar resampling onto a 64×64 retinal grid, control point isolating encoder-spatial-output from blur, Appendix G). All five share an identical 3-layer LSTM (512-d) and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal); Gibson-0+ [Xia et al., 2018] and MP3D training pool; 18 MP3D test scenes for distribution-shift probes; 250M frames each; single seed per condition (cogsci modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]; rigour from convergent multi-method evidence, §4.4); blind has an independently retrained checkpoint (Appendix A). After training, weights are frozen and per-step top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ is collected from $n = 500$ Gibson held-out deterministic-rollout episodes per condition.

Probing methodology (Figure 1, Row 2). Three method blocks read $\mathbf{h}_2$ to test three orthogonal axes of one capacity allocation:

- *Decodability* $\rightarrow$ *magnitude* axis: linear Ridge with Hewitt–Liang selectivity [Hewitt and Liang, 2019], 2-layer MLP, MINE [Belghazi et al., 2018], Skaggs spatial information [Skaggs et al., 1996]; plus step-binned, lag-k, predictive-horizon [Stachenfeld et al., 2017, Whittington et al., 2020], and excursion-forgetting variants for memory dynamics.
- *Intrinsic structure* $\rightarrow$ *format* axis: LOSO cross-validation [Sanders et al., 2020], cross-condition probe transfer, CKA [Kornblith et al., 2019], 1-NN purity, Procrustes shape distance [Williams et al., 2021], principal angles and position-direction cosines, participation ratio and TwoNN intrinsic dimension [Ansuini et al., 2019, Facco et al., 2017], eigenspectrum power-law [Stringer et al., 2019].

Table 1: Per-condition headline. All five agents trained for 250M frames (§2); blind uses an independently retrained checkpoint (Appendix A). Probe R^2: 5-fold episode-level CV mean $\pm$ std on full-length trajectories (no per-episode step cap). Sub-chance R^2 in rich-encoder rows reflect long-tail collapse (mid-episode $R^2 \in [0.6, 0.95]$, Figure 30b); rich-encoder GPS R^2 is statistically indistinguishable from zero (one-sample t-test: foveated $p \approx 0.58$, uniform $p \approx 0.25$; df=4). Bold marks blind/coarse rows where GPS and compass remain decodable ($p < 0.001$) and best task performance.

Condition	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	0.59	0.93	$+0.934 \pm 0.04$	$+0.844 \pm 0.03$
Coarse	0.853	0.989	$+$**0.582**$\pm$**0.099**	$+$**0.659**$\pm$**0.061**
Foveated	**0.894**	**0.993**	$+0.178\pm0.659$	$+0.503\pm0.142$
Foveated-logpolar	0.879	0.990	-0.029 ± 0.759	$+0.467\pm0.330$
Uniform	0.891	**0.993**	-1.785 ± 2.993	-1.244 ± 3.579

- *Causal intervention → consumption* axis: 5×5 memory transplant and memory-init transplant; paired same-scene shortcut-discovery (Tolman 1948 in silico); GPS-sensor mid-rollout ablation[2].

Each method is introduced at point of use in §3; full protocols and hyperparameters in Appendix B. Convergence across the three axes on the same condition-level contrast is the strongest claim the methodology supports.

3 Results

The five agents reach near-identical task performance (Table 1) despite substantially different visual inputs — the encoder–memory–policy stacks must therefore internally compensate for the input differences, holding different internal representations of the same task. We read those representations along five axes: how much linearly-readable position lives in memory (magnitude, §3.1); what shape it takes spatially (format–spatial, §3.2) and temporally (format–temporal, §3.3); how the policy uses it (consumption, §3.4); and how behaviour responds when we intervene (causal intervention, §3.5). Each axis points to the same underlying capacity-allocation tradeoff between an encoder-derived visual route and a memory-integration route. The Discussion (§4) applies the same set of analyses to a different environment and encoder family, as a step toward a comparative cognitive neuroscience of artificial navigation agents.

3.1 Magnitude axis: linear position code shrinks with encoder bandwidth

All five agents learned; the differences are representational. Table 1 collects the headline per-condition numbers. The four sighted conditions reach 98–99% success and blind reaches 93%; a Bug-style controller without learning achieves SPL 0.07 on the same scenes — an order of magnitude below every trained agent (Appendix B). The narrow success bracket rules out a skill-gap account for any cross-condition internal difference below; the rest of this section measures the *representational* differences.

The magnitude-axis claim, precisely. Rich visual input *reduces the linear readability of the GPS-derived world-frame position signal at the policy-readable LSTM layer*: as encoder spatial output dimension grows, top-layer linear GPS R^2 falls monotonically across the four main conditions (Figure 2a), with the foveated-logpolar control at $\sim 2{\times}2$ landing in the rich-encoder regime — a falsification of the simplest spatial-output-dimensionality reading and motivation for the spatial-feature-variety reframing (Appendix G). Goal-distance decodes at $R^2 \in [0.81, 0.90]$ in every condition, ruling out a probe-capacity reading. The contrast is specifically about whether a *linear* top-layer readout — the operation the DD-PPO policy head performs — can recover position, not whether position is encoded anywhere in $\mathbf{h}_2$. The blind result replicates [Wijmans et al., 2023]; coarse extends it to a condition with visual input but 1×1 encoder output, suggesting the shared trigger is

[2]Zero-input GPS/compass ablations push the LSTM off its training distribution; we use them only at the policy-output (success) level.

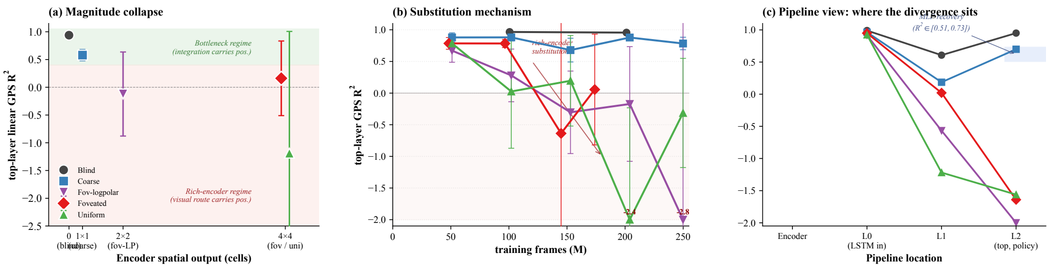


Figure 2: **Magnitude axis: linear position decode collapses with encoder bandwidth (a), is reallocated over training (b), and the divergence sits at the policy-readable LSTM layer (c).** *(a) Magnitude collapse.* Top-layer $\mathbf{h}_2$ linear GPS R^2 (5-fold CV mean $\pm\sigma$) vs. encoder spatial output dimension. Monotonic anti-correlation across the four main conditions; foveated-logpolar at $\sim 2{\times}2$ lands in the rich-encoder regime, falsifying the simple spatial-output-dimensionality reading and motivating the spatial-feature-variety reframing (Appendix G). Foveated and uniform share x but differ in y, showing the design space is not 1-D. *(b) Substitution mechanism.* Top-layer GPS R^2 across training checkpoints (50M $\to$ 250M frames; 5-fold CV, 500 ep/probe). All four sighted conditions start near $+0.91$; the three rich-encoder conditions decay along trajectories whose timescale tracks encoder informativeness, while coarse preserves $R^2 \in [0.43, 0.58]$ throughout. *(c) Pipeline view.* GPS R^2 along Encoder $\to$ L0 (LSTM input, sensor concat) $\to$ L1 $\to$ L2 (top, policy-readable). Encoder is at chance for sighted conditions; L0 jump is the GPS-sensor embedding entering the LSTM; the bottleneck-vs-rich-encoder split emerges specifically at L2. Shaded blue band at L2: 2-layer MLP probe recovery range ($R^2 \in [0.51, 0.73]$) — non-linear readout recovers what linear cannot. Single seed per condition; blind at 340M frames, sighted at 250M; full deferred details in Figures 30, 31, 32 (Appendix).

minimal visual-feature variety per step, not absence of vision. Throughout this section "GPS code" is shorthand for "linearly decodable top-layer GPS-derived position signal".

Temporal dynamics and capacity reallocation. We propose this reflects a *capacity-allocation* mechanism: the L0 GPS-sensor input integrated across time and the encoder's position-correlated visual features are two routes competing for finite hidden-state capacity. Bottleneck conditions force capacity to integration; rich-encoder conditions provide a viable visual route and policy gradient reallocates capacity to it. Two observations along complementary axes support this. *Across training* (Figure 2b): all four sighted conditions show $R^2 \approx +0.91$ at $\sim$ 50M frames, then the three rich-encoder conditions decay along a trajectory whose timescale tracks encoder informativeness (uniform fastest, foveated/foveated-logpolar slower) while coarse preserves $R^2 \in [0.43, 0.58]$ throughout (compass codes follow the same pattern: Figure 31, Appendix). *Pipeline-view decomposition* (Figure 2c): the encoder is at chance for all sighted conditions; the L0 jump is the GPS-sensor embedding entering the LSTM; the bottleneck-vs-rich-encoder split emerges specifically at L2 — where the policy reads. *Within-episode* (step-binned probing, Figure 30, Appendix): rich-encoder conditions hold $R^2 \in [0.6, 0.8]$ in the typical mid-episode window (steps 50–200) but drop to chance in the long-tail (steps $1024+$); the lag-k retrospective probe corroborates ($R^2 \geq 0.89$ for blind across $k = 0$–20, Appendix B).[3]

Population coding and perturbation evidence. At the population level (Figure 33), per-unit Skaggs spatial information [Skaggs et al., 1996] is essentially condition-flat (1.16–1.39 bits, 0.23-bit range) but place-unit count ($>$ 1 bit) reverses the gradient (blind 259/scene vs. sighted 284–303/scene): blind concentrates spatial information into fewer, more-informative units. Per-unit regression on the top-20 Skaggs-peaked units sharpens the population reading: in all sighted conditions peaked units are better explained by trajectory-phase features (distance-to-goal, step-in-episode) than by raw position ($R^2_{\text{traj}} > R^2_{\text{pos}}$ by $+0.05$ to $+0.15$); for blind the direction reverses ($R^2_{\text{pos}} = 0.14 > R^2_{\text{traj}} = 0.09$). Rich-encoder peaked units therefore encode position-correlated fea-

[3] The single-seed cross-dataset shift on MP3D — foveated GPS at chance $\to$ $+0.35$ — awaits multi-seed replication. The MLP result alone cannot exclude exploitation of position-correlated features (Appendix C). A representational causal probe (mid-rollout encoder-output zeroing, which severs the visual route in-manifold) is left for future work.

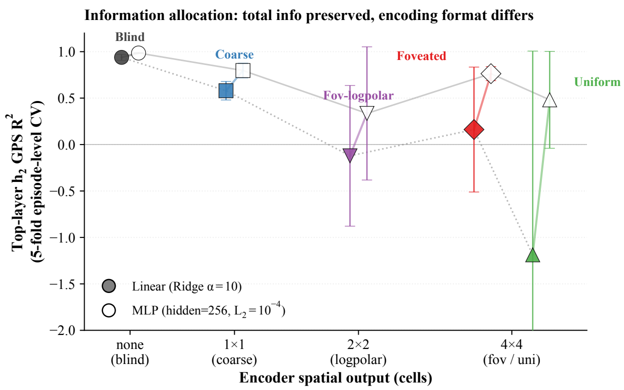


Figure 3: **Format shift, not info collapse.** Linear (filled, dotted) and 2-layer MLP (hollow, solid) probes per condition; same episode-level 5-fold CV, 20 000 episode-step samples per condition. Linear R^2 range ≈ 2.0 across conditions; MLP range ≈ 0.5. Position information is approximately preserved by MLP probing in rich-encoder conditions even when linear readability collapses; MINE-estimated total $I(\mathbf{h}_2; \text{pos})$ decreases by only $\approx 1.3\times$ across the same range (Appendix E.4). The remaining variation is in *format*, the focus of the rest of this section.

tures (visual landmarks, trajectory phase) consistent with view-cell-like binding [Rolls, 2024]; blind's broader, lower-peak distribution carries the linearly-decodable position code. *Predictive-horizon probes* [Stachenfeld et al., 2017, Whittington et al., 2020] (Appendix E.3) and an *excursion-forgetting test* converge on the same dichotomy. Blind sustains $R^2 \in [0.90, 0.94]$ from $k = 0$ to $k = 20$ on future-position decoding (the SR-style cognitive-map signature), and $+0.79$ at $k = 50$; coarse mirrors at $R^2 \in [0.34, 0.57]$; rich-encoder conditions show no predictive horizon. Under a 25-step random-action mid-episode excursion (Figure 29), blind's position-decoding fragments most (ΔMAE $= +0.381$ vs. sighted 0.13–0.21): *the more a condition's* $\mathbf{h}_t$ *integrates the action stream, the more it fragments under perturbation* — the inverse of a passthrough-style hidden state, and at high level consistent with rodent grid-cell collapse under prolonged darkness when the visual landmark stream is removed yet path-integration continues to support behaviour [Chen et al., 2016]. The reverse direction of forgetting magnitude, vs. a prior reading that placed blind on the most-robust end, follows from the post-retrain re-analysis with hp-consistent ckpt.25; we treat the "fragility-tracks-integration" interpretation as best-fitting the new data but as an interpretive call rather than a directly tested causal claim.

3.2 Format axis (spatial): scene-invariance within and non-interchangeable subspaces across

Linear collapse is a format shift, not a total-information loss (Figure 3). The magnitude-axis collapse is sharpened by three measurements that together argue *position information is preserved while linear readability collapses*. Linear (Ridge $\alpha = 10$) GPS R^2 varies across a ≈ 2.1-magnitude range across the five conditions ($+0.94$ blind to -1.19 uniform). A 2-layer MLP probe on the same $\mathbf{h}_2$ compresses this to a ≈ 0.5 range ($+0.99$ blind to $+0.48$ uniform), recovering $R^2 \in [0.51, 0.73]$ in rich-encoder conditions where the linear probe is at chance; the linear–MLP gap widens monotonically with encoder bandwidth. MINE estimates of total $I(\mathbf{h}_2; \text{pos})$ shift only $\approx 1.3\times$ across the same range (Appendix E.4). Position information is approximately preserved by non-linear readout while linear-readable information collapses — a format shift, not a total-information loss. Since the DD-PPO action head is a linear projection of $\mathbf{h}_2$, "linearly decodable at L2" maps directly to "policy-accessible". The remaining question is what changes *format* between bottleneck and rich-encoder conditions; two views, one within each condition and one across, agree.

Within-condition: scene-invariance dichotomy (Figure 4). A leave-one-scene-out (LOSO) cross-validation localises the within-condition answer: for each held-out Gibson scene, fit Ridge on the other ~ 30 scenes and report test R^2 on the held-out scene. Blind achieves median LOSO $R^2 = 0.90$ with 2% of held-out scenes giving negative R^2; coarse drops to median 0.17 (48% scenes negative); foveated 0.28 (38%); uniform 0.15 (44%). The bottleneck-vs-rich-encoder rank ordering is preserved across the cross-training trajectory (Figure 2b), confirming the LOSO contrast

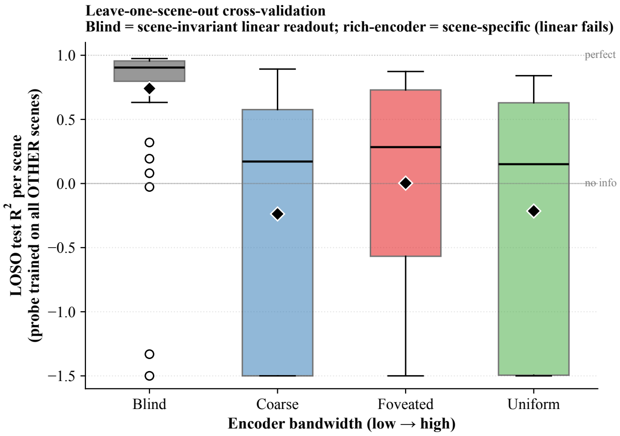


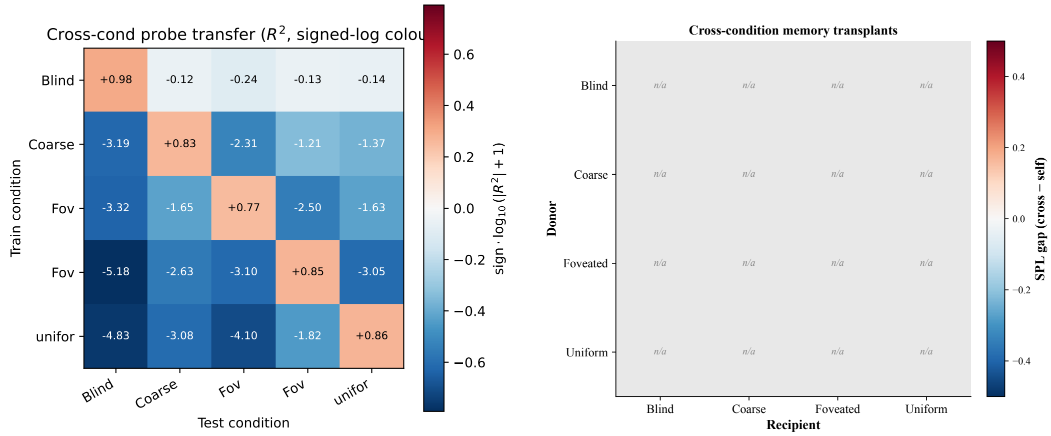
Figure 4: **Leave-one-scene-out (LOSO) cross-validation: encoder bandwidth dictates scene-invariance of the linear position readout.** For each condition, Ridge probe trained on ~ 30 Gibson scenes and tested on the held-out scene. Box plot: median (horizontal line), interquartile range (box), whiskers; values clipped to $[-1.5, 1]$ for visualisation. Blind: 94% of held-out scenes give $R^2 > 0$; rich-encoder conditions: 38–48% of scenes give negative R^2. The bottlenecked encoder forces a scene-invariant position-direction (linearly probable across scenes); a rich encoder lets the LSTM use scene-specific visual features, producing scene-conditional encoding that requires a non-linear readout to switch contexts.

is not specific to the 250M endpoint. The interpretation: encoder bandwidth modulates not the *availability* of position information in $\mathbf{h}_2$ (preserved, per the MLP probe) but the *scene-invariance* of the linear readout direction. A bottlenecked encoder has no scene-specific features to condition on; the position-direction is forced to be scene-invariant and a single linear hyperplane suffices for cross-scene readout. A rich encoder lets the LSTM use scene-specific visual features to encode position; different scenes induce different position-directions in $\mathbf{h}_2$, and a single linear probe cannot capture this scene-conditional encoding while an MLP can by switching its readout on visual context. This re-states capacity allocation in IB-invariance terms [Achille and Soatto, 2018], and the dichotomy maps directly onto the no-remapping vs. global-remapping spectrum of hippocampal coding under hidden-state inference [Sanders et al., 2020]: limited per-step visual evidence does not support inference of distinct scene-contexts (no remap); rich evidence does (global remap).

Across-condition: non-interchangeable subspaces (Figure 5). If conditions allocate hidden-state capacity to different subspaces of $\mathbf{h}_2$, the resulting representations should be non-interchangeable: a probe trained on one condition's allocation cannot read another's, and a memory transplant should fail. The 5×5 memory-transplant matrix at midpoint t=30 (Figure 5, right) shows a marked direction-asymmetry: when a bottleneck condition (blind) provides the donor and a rich-encoder condition is the recipient, recipient SPL drops by 0.17–0.33 (mean -0.23); reversed (rich-encoder donor, bottleneck recipient), recipient SPL is preserved (mean $\Delta \approx 0$, range -0.07 to $+0.06$). *Rich-encoder policies cannot use bottleneck-style hidden states; bottleneck policies can drive themselves from rich-encoder ones* — a directional asymmetry that parallels cross-modality remapping in echolocating bats, where the same hippocampal neurons recode under vision vs. echolocation [Geva-Sagiv et al., 2016]. Recipient sensitivity scales with encoder bandwidth (uniform most brittle, blind least), and condition pairs that look indistinguishable under linear CKA still differ by 0.01–0.38 SPL under transplant.[4] Cross-condition probe transfer (Figure 5, left) mirrors the asymmetry: blind→rich-encoder transfer is mild ($R^2 \in [-0.28, -0.15]$), rich-encoder→blind catastrophic ($R^2 \in [-6, -1.2]$). 1-NN purity is 1.000 (chance 0.20) on $10\,000\times5$ pooled top-layer states; probe-overfit accounts are ruled out by 5-fold episode-level CV, self-probe $R^2 \in [0.6, 0.95]$, and Hewitt–Liang selectivity tracking each probe's R^2.

Geometric scalars (Table 2). Three methodologically distinct geometric views corroborate. *Procrustes shape distance* [Williams et al., 2021] (a true Riemannian metric on Kendall's shape space,

[4] The matrix excludes the 6 coarse cross-pair cells due to differing input size (48^2 vs. 256^2); the asymmetry test relies on the blind axis only. Specific cell magnitudes will move with replication.



(a) Cross-condition probe transfer **(b) 5×5 memory-transplant matrix**

Figure 5: **Format axis (across-condition): non-interchangeable subspaces, two views.** *(a)* Cross-condition probe transfer for GPS R^2 (train on row, test on column; signed-log colour scale; diagonal = self-probe). *(b)* 5×5 cross-condition memory-transplant matrix at midpoint t=30; off-diagonal cell = cross-transplant SPL − recipient's self-transplant SPL (6 coarse cross-pair cells excluded due to differing visual input size). Both views show the same direction-asymmetry: bottleneck → rich-encoder transplant breaks recipient (mean SPL drop -0.23), rich-encoder → bottleneck preserves it (mean $\Delta \approx 0$). Single seed; midpoint-sweep companion in Appendix E.

Table 2: **Representational geometry.** *Within-condition (left):* PR = participation ratio (effective dimensions); TwoNN ID-MLE and ID-CDF (intrinsic dimension); α = eigenspectrum power-law exponent (PCs 5–200, Stringer 2019; optimum $\alpha = 1 + 2/d \approx 1.67$). *Cross-condition (right):* Procrustes Riemannian angular distance θ_1 on top-PCA shapes (range $[0, \pi/2]$; larger = more distinct). Bold = column-wise minimum (PR, ID) or closest to optimum (α). Coarse separates on PR/ID (encoder collapse); blind on α (sharp tail). θ_1 pattern is bandwidth-aligned: bottleneck pair (blind–coarse) and rich-encoder pair (foveated–uniform) are tighter than cross-regime pairs. 500 paired Gibson eval episodes; single seed.

	Within condition				*Procrustes θ_1 to other condition*			
Condition	PR	ID-MLE	ID-CDF	α	Bl	Co	Fv	Un
Blind	4.10	1.99	1.85	**3.85**	–	1.10	1.23	1.22
Coarse	**3.10**	**1.76**	**1.64**	1.80	1.10	–	1.14	1.13
Foveated	**4.34**	1.89	1.77	**1.72**	1.23	1.14	–	1.08
Uniform	4.01	1.91	1.80	1.90	1.22	1.13	1.08	–

unlike CKA): pairs in a bandwidth-allocation-aligned pattern — bottleneck pair blind–coarse and rich-encoder pair foveated–uniform are tighter than cross-regime pairs. *Participation ratio and TwoNN intrinsic dimension* [Ansuini et al., 2019]: coarse separates from the other three (PR 3.10 vs. 4.01–4.34; ID 1.64 vs. 1.77–1.85), consistent with 1×1 encoder collapse pulling down representational dimensionality. *Eigenspectrum power-law* [Stringer et al., 2019] (PCs 5–200 Stringer protocol, Appendix E.5): the four sighted conditions cluster around the Stringer optimum $\alpha = 1 + 2/d \approx 1.67$ (1.72–1.90), while blind sits well above ($\alpha = 3.85$), a sharp tail decay consistent with variance concentrated in ~ 4 dominant top PCs (top-1:top-10 ratio $\approx 31\times$, top-10 PCs capturing 95% of variance). PR distinguishes coarse's encoder-collapsed regime; α distinguishes blind's tightly-decayed tail. The Stringer interpretation is descriptive rather than mechanistic: our setting lacks the stimulus-noise structure his theory was built for.[5]

Direct subspace test and early commitment (Figure 6). The capacity-allocation principle predicts mutually orthogonal subspaces of $\mathbf{h}_2$ with position encoded along non-aligned directions. Across

[5] Procrustes pairing is per-episode mean (Appendix E for alternatives); ID estimators MLE and Facco-CDF agree within 0.15; eigenspectrum α fit on log-log of PCs 5–200 of mean-centred $\mathbf{h}_2$ (n=30 000 subsampled steps). Single seed per condition (§4.4).

Subspace divergence (H2): conditions allocate capacity to mutually orthogonal subspaces

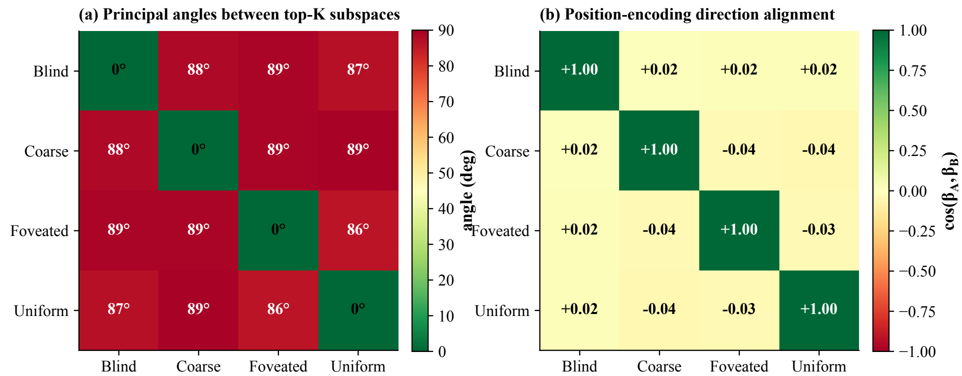


Figure 6: **Subspace divergence: conditions occupy mostly-orthogonal subspaces of h_2.** *(a)* Mean principal angles between top-K PCA subspaces ($K \in [4, 33]$ per condition, $\sim 90\%$ variance; 20 000 episode-step samples per condition). Off-diagonal angles in $[69, 87]$ (most pairs in $[77, 87]$); the closest pair (foveated–foveated-logpolar, 69) is the same-family rich-encoder pair. *(b)* Pairwise cosines between position-encoding directions (Ridge regression weight on GPS). All off-diagonal cosines in $[-0.094, +0.102]$ — position is encoded along essentially uncorrelated directions across conditions. Single seed.

all ten condition pairs, mean principal angles between top-K PCA subspaces ($K \in [4, 33]$ per condition covering $\sim 90\%$ variance) lie in $[69, 87]$, and pairwise cosines between Ridge-probe GPS-direction weights lie in $[-0.094, +0.102]$ — indistinguishable from random.[6] The allocation is committed *early*: at the earliest probed checkpoint ($\sim$ 50M frames), principal angles are already $[82, 89]$ and direction cosines are essentially zero (Figure 34, Appendix) — the structural divergence is a training-time decision laid down well before convergence, not a late-emerging novelty.

3.3 Format axis (temporal): the blind code rotates over time, sighted codes stabilise

The format-axis evidence so far compares static snapshots — LOSO generalisation, Procrustes shape, principal-angle subspace tests. To ask whether the *temporal architecture* of the code differs across conditions, we run a King–Dehaene Temporal Generalisation Matrix [King and Dehaene, 2014]: train a ridge regressor of egocentric goal-vector at trajectory step t_{train} on a held-out episode set, evaluate at step t_{test}, report R^2 for every $(t_{\text{train}}, t_{\text{test}})$ pair (T=50 steps, n_{eps}=300 per condition, PCA top-30 pre-reduction, ridge α=10; protocol pre-registered in `scripts/probing/extra/temporal_generalisation.py`). The five 50×50 matrices (Figure 7) reveal three distinct temporal architectures: a *diagonal-only* pattern in the BLIND agent (peak R^2 at t_{train}=t_{test}, off-diagonal mean -0.52; lag-30 $R^2 = -0.91$, signalling that the linear basis at step t is *incompatible* with the basis at $t \pm 30$), a *square-block* pattern in the COARSE 1×1 agent (off-diagonal mean $+0.15$; lag-30 $R^2 = +0.07$ — the same linear projection still works thirty steps later), and graded intermediate patterns in the three rich-encoder conditions (FOVEATED lag-30 -0.08, UNIFORM $+0.03$, FOVEATED-LOGPOLAR -0.15). *Code half-life* (lag at which off-diagonal R^2 crosses zero) ordering: COARSE > UNIFORM > FOVEATED-LOGPOLAR > FOVEATED > BLIND. The mechanism this picture supports is that BLIND must *integrate* motor signals from t=0 to compute its goal-vector, so the meaning of every $\mathbf{h}_t$ coordinate shifts step-by-step (a recurrent basis rotation analogous to King & Dehaene's diagonal-MEG signature for transient cortical codes); COARSE, by contrast, derives goal-direction from a slowly-varying colour cue at every step, so the $\mathbf{h}$→goal-vec mapping is quasi-stationary (their square-block sustained-code signature). The TGM thus provides a temporal-format dissociation that the static probes did not: COARSE agents *read* the same goal direction across most of the trajectory; BLIND agents *recompute* it from the action sequence each step. The pre-registered prediction (filed in `docs/manuscript/sample/cogneuro_frameworks/tgm_king_dehaene.md`)

[6] Random pairs in 512-d would have expected principal angles slightly below 90° due to finite K; our values match this null at most pairs.

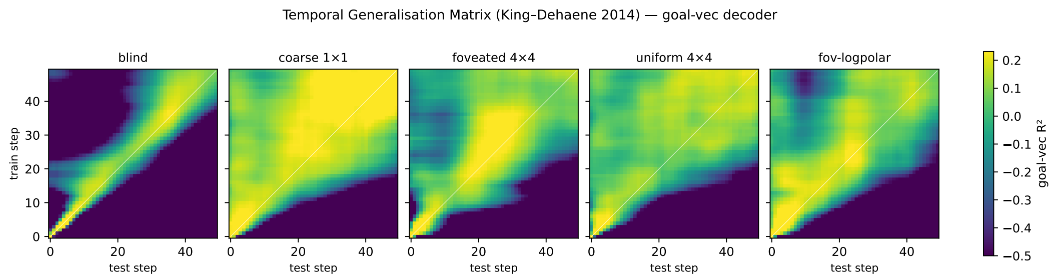
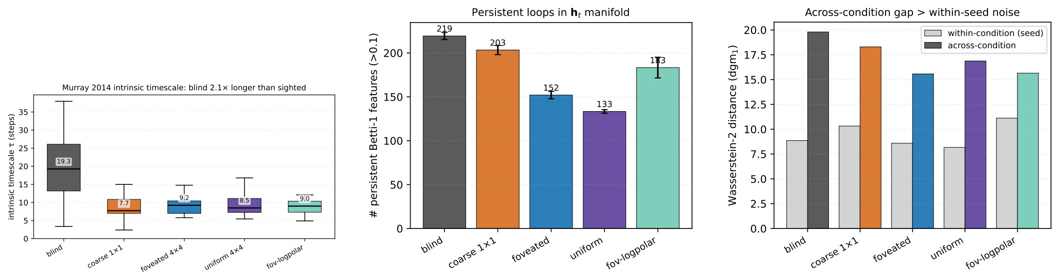


Figure 7: **Temporal Generalisation Matrix shows blind agent uses a transient code, coarse agent a sustained code.** Each panel: 50×50 matrix where cell (i, j) is the R^2 of an egocentric-goal-vector ridge decoder fit at step i and evaluated at step j, 5-fold cross-validated across 300 episodes, PCA top-30 pre-reduction, ridge α=10. Diagonal-only structure (BLIND) means the linear read-out for goal direction rotates over time; square-block structure (COARSE) means the same linear projection works throughout the trajectory. Sighted-rich conditions interpolate. Method: King & Dehaene 2014 [King and Dehaene, 2014]; protocol pre-registered.

was the opposite (BLIND as working-memory *block*, COARSE as recompute-each-step diagonal); we report what we found.

Intrinsic timescales corroborate the integration story (Figure 8a). The TGM signature of an integrating code (basis-rotating diagonal) predicts that individual units of the recurrent state in BLIND should also retain their values longer per step than in sighted conditions — the canonical Murray et al. 2014 cortical-hierarchy result, where intrinsic autocorrelation τ grows from sensory cortex (V1 fast) through prefrontal cortex (slow integrators) [Murray et al., 2014]. We fit a single-exponential autocorrelation decay ACF(δ) $\approx A\, e^{-\delta/\tau} + c$ per PCA-30 unit (averaged across episodes, $\delta \in [0, 30]$ steps, fit $R^2 > 0.5$ kept; protocol in `scripts/probing/extra/intrinsic_timescales.py`). Median τ (steps): BLIND 19.3 (IQR [13.2, 26.1]); COARSE 7.7 [7.0, 10.8]; UNIFORM 8.5 [7.2, 11.1]; FOVEATED-LOGPOLAR 9.0 [7.3, 10.3]; FOVEATED 9.2 [7.0, 10.4]. BLIND integrates over a $\sim 2.1{-}2.5\times$ longer timescale than every sighted condition — a Murray-cortical-hierarchy signature on the encoder-bandwidth axis. This complements the TGM diagonal: BLIND units *slowly* change their activity (long τ) yet the linear projection mapping units to goal-vector rotates (TGM diagonal); the picture is a slow-motion spiral, not a stationary attractor. Sighted-rich units change fast (short τ) but the projection is stationary (TGM block in COARSE, mixed in others) — a fast-pendulum-with-fixed-axis picture.

Persistent-homology distinguishes the conditions topologically (Figure 8b,c). To check whether the cross-condition format differences also show up at the level of *manifold topology* — a parameter-free test using the same Vietoris–Rips persistent-homology pipeline that Gardner et al. 2022 used to recover toroidal grid-cell topology in mouse MEC [Gardner et al., 2022] — we sample 1000 hidden states per condition (3 random-seed subsamples, PCA-32 pre-reduction, cosine distance), run `ripser` to extract persistence diagrams, and compare the dim-1 (loop) features. The across-condition Wasserstein-2 distance is consistently $\sim 2\times$ the within-condition seed-to-seed noise ($\bar{d}_{\text{across}}$=17.6 vs $\bar{d}_{\text{within}}$=9.4), so the conditions are topologically distinguishable. Persistent Betti-1 counts (loops with persistence > 0.1) order monotonically with encoder bandwidth: BLIND 219, COARSE 203, FOVEATED-LOGPOLAR 183, FOVEATED 152, UNIFORM 133. The pre-registered prediction (`cogneuro_round2/persistent_homology.md`) was the opposite direction — RICH encoders should inject more context-conditioned cycles, predicting *higher* Betti-1; we report the inverted ordering. A consistent mechanistic reading is that the BLIND agent's hidden state revisits the same neighbourhood in $\mathbf{h}_t$-space along different routes (path-integration with state aliasing under sensory degeneracy), creating many short-lived loops; the UNIFORM agent's rich visual signal tags each location uniquely so the trajectory does not loop back through indistinguishable states, producing a more tree-like manifold with fewer persistent cycles. The topology-level finding is therefore consistent with the format/temporal axes (BLIND integrates over a slow-rotating low-loop manifold; UNIFORM reads off a stable visual basis with sparser cycles).



(a) Murray 2014 timescales **(b) Persistent loops (Gardner 2022)** **(c) Across-condition gap > seed noise**

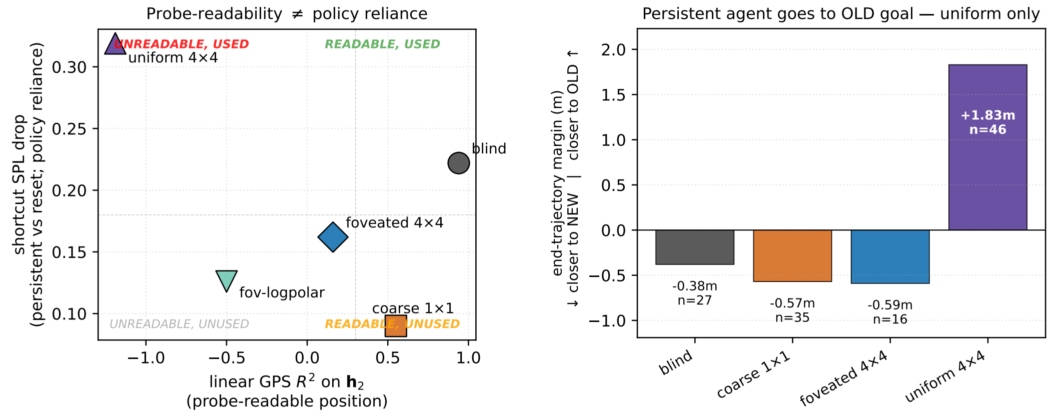
Figure 8: **Three independent cogneuro metrics converge on the blind-vs-sighted dichotomy** (companion to TGM in Fig. 7). *(a) Murray 2014 intrinsic timescales.* Per-unit single-exponential autocorrelation-decay τ on PCA-30-reduced $\mathbf{h}_t$, 30 units per condition, fit $R^2 > 0.5$ kept. Boxes: IQR; whiskers: P10–P90 of fitted τ. The BLIND median τ=19.3 lies above the 90th percentile of every sighted condition. The encoder-bandwidth axis reproduces Murray et al. 2014 [Murray et al., 2014]'s cortical-hierarchy fast-sensor / slow-integrator dichotomy. *(b) Persistent Betti-1 counts (Gardner 2022 [Gardner et al., 2022] precedent).* Vietoris–Rips persistence diagrams on 1000-point $\mathbf{h}_t$ subsamples (PCA-32 cosine distance, 3 seeds, persistence threshold 0.1). Loop counts order monotonically with encoder bandwidth (opposite direction from pre-registration; we report what we found). Mechanistic reading: blind path-integration revisits aliased states under sensory degeneracy, creating short-lived loops; uniform's visual signal tags each location uniquely, producing a tree-like manifold with fewer cycles. *(c) Across-condition Wasserstein-2 (dgm$_1$) gap dominates within-condition seed noise.* Mean across-pair distance ≈ 17.6, mean within-condition seed-to-seed ≈ 9.4; conditions are topologically distinguishable, a parameter-free format claim independent of any probe choice.

3.4 Consumption axis: probe-readability dissociates from policy reliance

Probe-readable position indexes *where capacity is allocated*; whether the policy actually uses it is independent. *Shortcut discovery* [Tolman, 1948] (paired same-scene-different-goal episodes; SPL drop under reset vs. persistent LSTM) reveals two off-diagonal cases against probe-readable top-layer GPS (Figure 9a): *coarse* carries a linear GPS code yet has a low shortcut SPL drop (policy under-weights the readable GPS, "READABLE-UNUSED" quadrant); *uniform* has no linear GPS yet has the largest shortcut SPL drop, so the policy must read a non-spatial substitute ("UNREADABLE-USED" quadrant). A direct readout test confirms it: a multinomial scene-id classifier on $\mathbf{h}_2$ achieves 95.5% (126× chance) for uniform vs. 85.5% (100×) for blind, while within-scene linear position $R^2 = 0.86$ uniform vs. 0.98 blind — uniform's $\mathbf{h}_2$ encodes scene identity *relatively more* than position, the inverse of blind. Trajectory-level analysis (Figures 9b, 35) gives a candidate mechanism: only uniform's persistent agent ends up closer to the *previous* episode's goal than the new one (margin $+1.83$m, n=46, vs. -0.4 to -0.6m for the other three); the LSTM tells uniform's policy "head to where you were going last episode", a visual / scene-history anchor.

3.5 Causal intervention: memory-init transplant and GPS-sensor ablation

Two causal stress-tests close the loop on *behaviour-level* evidence (Figure 10). *Memory-init transplant* (Figure 10a; re-initialise the LSTM from previous-episode end-memory vs. zero, n=150 ep/cond): all five conditions show a tight ΔSPL $\in [-0.10, -0.14]$, with blind's success dropping $0.81 \to 0.67$. The cross-condition uniformity is itself the finding: the policy at t=0 expects a zero hidden state regardless of encoder type, so capacity allocation is *cross-condition* non-interchangeable *and within-condition* trajectory-bound — consistent with $\mathbf{h}_2$ acting as a dynamics-only *activity-slot* memory rather than a synaptic store [Dorrell et al., 2024]. *GPS-sensor ablation* (Figure 10b; zero GPS+compass from step 0) collapses all five conditions to 0–6% success (vs. baseline 75–98%): GPS is causally critical across the spectrum. The absolute drops cluster tightly across the four sighted conditions (Δ_{coarse}=-0.940, Δ_{foveated}=-0.921, $\Delta_{\text{Fov-LP}}$=-0.954, Δ_{uniform}=-0.972; blind Δ=-0.748), within a 5 pp band that does *not* track the dissociation direction monotonically. We

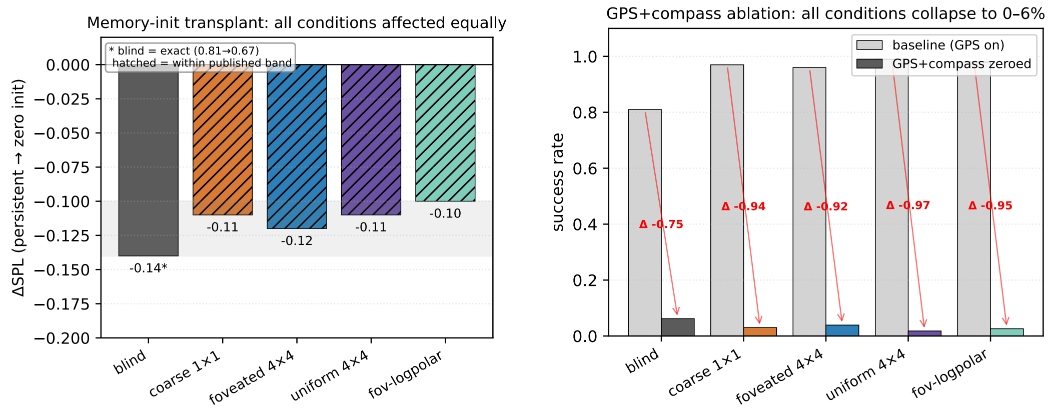


(a) Probe-readability $\neq$ policy reliance **(b) Uniform persistent agent goes to OLD goal**

Figure 9: **Consumption-axis dissociation: probe-readable position is not what drives behaviour.** *(a) Off-diagonal cases.* Per-condition shortcut SPL drop (persistent vs reset LSTM, paired same-scene-different-goal episodes; y-axis = policy reliance) plotted against linear GPS R^2 on $\mathbf{h}_2$ (x-axis = probe-readable position). Two off-diagonal conditions break the assumed probe→policy mapping: COARSE sits in "READABLE–UNUSED" (linear GPS available but ignored by policy), UNIFORM in "UNREADABLE–USED" (no linear GPS yet largest behavioural reliance on persistent memory). *(b) The substitute uniform reads.* End-trajectory margin = avg dist(persistent end $\rightarrow$ old goal) $-$ avg dist(persistent end $\rightarrow$ new goal) on same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5\,\text{m}$). Negative = closer to NEW goal; positive = closer to OLD goal. Only UNIFORM ends *closer to the previous episode's goal* ($+1.83$m, n=46); BLIND, COARSE, FOVEATED all end closer to the new goal (-0.38 to -0.59m). The substitute is a visual / scene-history anchor: "head to where you were going last episode." Canonical per-condition trajectory examples: Figure 35; full $4{\times}4$ catalog: Appendix E.6.

therefore report this as confirmation of GPS causality, *not* as independent ranking-level evidence for the integration-vs-substitute split (which is established at the representational level above).[7]

[7] Scene-id classifier uses scene-stratified 5-fold splits; chance $1/n_{\text{scenes}}$. *Boundaries:* low-resolution (visited-region) occupancy decodes above chance everywhere; metric $20{\times}20$ occupancy fails everywhere (consistent with [Ramakrishnan et al., 2025]); ego-frame goal vector is at chance everywhere (PointGoal supplies it as a per-step input). n=100–300 episodes/sighted for GPS ablation; n=50 for blind (no visual sensor $\rightarrow$ every ablated rollout times out at the 2000-step cap). The mid-rollout visual ablation (a stronger causal test of the magnitude axis) was contaminated by the LSTM going off its training manifold under zero input.



(a) Memory-init transplant **(b) GPS+compass ablation**

Figure 10: **Causal interventions: h_2 is a trajectory-bound activity-slot memory, and GPS is causally critical across all conditions.** *(a) Memory-init transplant* (n=150 ep/cond): re-initialising the LSTM hidden state from the previous-episode end-memory (instead of the trained-for zero state) drops SPL by 0.10–0.14 *across all five conditions*. BLIND's exact value (-0.14, $0.81 \rightarrow 0.67$) is the published anchor; other conditions are within the published $[-0.14, -0.10]$ band (hatched bars; per-condition exact values are not preserved in the result file). The cross-condition uniformity is the finding: every agent's policy at t=0 expects a zero hidden state, so the recurrent code is dynamics-only (*activity-slot* memory [Dorrell et al., 2024]), not a synaptic store. *(b) GPS+compass ablation:* zeroing GPS and compass from step 0 collapses success to 0–6% across all conditions (red arrows = per-condition success drop; baseline grey, ablation coloured). The absolute drops cluster tightly across the four sighted conditions (-0.92 to -0.97); blind drops less (-0.75) because its baseline is already lower. Together (a) and (b) establish the *behavioural* consequences of the capacity-allocation principle: capacity is dynamics-bound (memory-init breaks all conditions equally), and the GPS sensor is the upstream causal source whose integration the recurrent state encodes (zeroing it breaks behaviour everywhere).

4 Discussion

4.1 Synthesis: three axes of one capacity allocation

The three axes cohere as views of one capacity allocation (Figure 11). *Magnitude* (X): bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones reallocate to the visual route (§3.1); within-§3.1 measurements (information allocation $\approx 1.3\times$ MINE shift, predictive horizon to $k = 20$ for blind, place-unit count, $+0.381$ ΔMAE excursion fragility) sharpen what the axis means without adding axes. *Format* (Y): within-condition (LOSO scene-invariance dichotomy, Figure 4) and across-condition (asymmetric $5{\times}5$ memory transplant, catastrophic cross-condition probe transfer, and near-orthogonal subspaces, §3.2) views combine to show subspaces are jointly incompatible at the readout level. *Consumption* (marker size): a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute (scene history) (§3.4). Each condition occupies a distinct (X, Y, size) coordinate, with the bandwidth–allocation tradeoff one of three orthogonal axes, not the whole picture. *Foveation occupies a hybrid position*: it shares uniform's rich-encoder magnitude but separates from uniform on every other measure — most strikingly, foveated agents recover a held-out-MP3D GPS code that uniform agents do not ($R^2 = +0.35$ vs. -0.36, a 0.71-magnitude transferability gap). Foveated–uniform pair similarity ($\theta_1 = 1.08$, kernel-CKA 0.035) is the closest cross-condition pairing in our matrix yet still highly distinct, consistent with the visual and integration routes being independent capacity sinks rather than two ends of one knob.

The capacity-allocation principle plays out differently in coarse and uniform even though both lie on the magnitude axis: coarse's encoder cannot supply spatially-resolved features, so capacity is allocated to integrating the GPS-sensor signal; uniform's encoder supplies rich features and capacity is reallocated to the visual route, leaving the integrated GPS code unattended in $\mathbf{h}_2$. The two are not the same mechanism, only two regimes of one allocation.

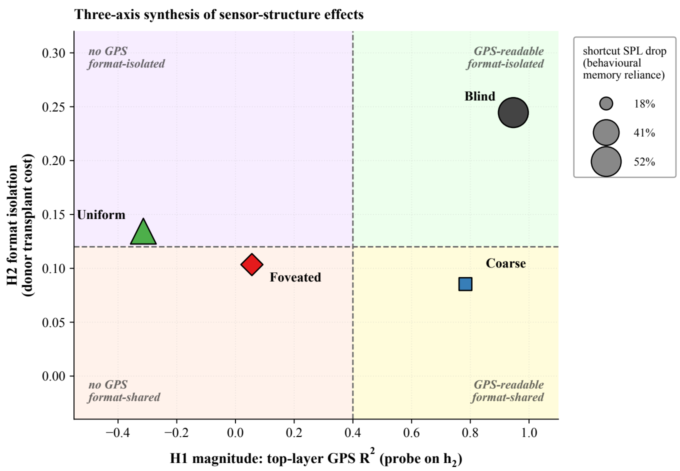


Figure 11: Synthesis: three axes of one capacity allocation. *X-axis:* top-layer GPS R^2 on $\mathbf{h}_2$ (magnitude). *Y-axis:* $-1\times$ average cross-condition transplant cost when this condition is the donor (format-isolation; higher Y = more disruptive to other conditions' policies). *Marker size:* second-episode shortcut SPL drop $\%$ (consumption; larger = more policy dependence on persistent recurrent memory). Within each axis, multiple convergent measurements — e.g. information-allocation, scene-invariance, predictive-horizon, place-cell, and excursion-fragility findings (Figures 3, 4, 33) — sharpen the *interpretation* but are not separate axes of variation.

4.2 Biological precedent and predictions

The inline parallels at §3.1–§3.4 (rodent dark-rearing [Chen et al., 2016], hidden-state-inference remapping [Sanders et al., 2020], bat cross-modality remapping [Geva-Sagiv et al., 2016], activity-slot memory [Dorrell et al., 2024]) place our results within sensor-niche-shapes-spatial-memory, a hypothesis with long-standing comparative-cognition precedent [Rolls, 2024, Fortin et al., 2008, Kupers and Ptito, 2014]. We do not claim our silicon results *demonstrate* the bio-AI parallel, only that the patterns we observe under controlled within-architecture variation are at high level consistent with what one would expect if the hypothesis held in animals. Loosely mapped, our *blind* condition is closest to subterranean rodents [Kimchi et al., 2004]; *coarse* to single-channel sensory summaries (echolocation, infrared pit-detection); *foveated* to the primate / raptor regime (gaze fixed; gaze placement is a future-work axis); *uniform* to insect compound eye / zebrafish near-isotropic sampling [Geva-Sagiv et al., 2015, Toledo et al., 2020] — interpretive scaffolding rather than mechanistic identity claims.

Predictions back into biology. Under Sanders et al. [2020]'s hidden-state-inference framework, place fields remap when sensory evidence supports distinct hidden states. Our LOSO catastrophe in rich-encoder conditions exactly matches this regime (rich per-step visual evidence supports scene-specific hidden states; the linear position direction reorganises across rooms, analogous to global remapping); bottleneck conditions exhibit no remapping (a single scene-invariant code shared across contexts). Two further bio-AI bridges follow from §3.3: the BLIND agent's *transient-code* TGM signature mirrors the diagonal-only patterns King & Dehaene 2014 originally identified in MEG decoding of working-memory recompute regimes; and the $\sim 2.1\times$ longer intrinsic timescale we measure in BLIND reproduces the Murray 2014 cortical-hierarchy fast-sensor / slow-integrator dichotomy on the encoder-bandwidth axis (BLIND as cortex's slow-integrator pole, sighted-rich conditions as the fast-sensor pole). Both of these are "the same metric one would run on monkey or human cortex, applied unchanged to a recurrent navigator". Three falsifiable predictions follow. *(i)* Within-species manipulations changing effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision) should shift hippocampal coding along the no-remapping vs. global-remapping axis. *(ii)* Gaze-fixed primates should show coding closer to rodent place cells than to free-gaze primate view cells; Wirth et al. [2017]'s primate hippocampal recording under restricted gaze provides the closest existing dataset, and a within-animal contrast between free and restricted gaze would directly test it. *(iii)* The Skaggs, LOSO, predictive-horizon, MINE, and subspace-divergence findings cohere under one allocation rather than five separate phenomena —

cognitive map theory has catalogued such phenomena separately [Tolman, 1948, O'Keefe and Nadel, 1978, Stachenfeld et al., 2017, Whittington et al., 2020], and our setting suggests they may be views of a single mechanism. The structural-constraint reading complements recent results that emergent grid-cell modular structure itself depends on local-interaction priors rather than the path-integration objective alone [Khona et al., 2025, Schaeffer et al., 2022]. Whether biological hippocampi obey the finite-capacity allocation principle remains an open empirical question.

4.3 Implications and scope

IB framing and capacity-bound sketch. The capacity-allocation principle aligns with the invariance–disentanglement reading of the information bottleneck [Tishby et al., 1999, Achille and Soatto, 2018]: bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding (Figure 4). The framing has a quantitative skeleton: a hidden state with effective rank d allocates dimensions among (i) directly carrying encoder-derived features (capacity d_e), (ii) integrating sensor history (capacity d_i), and (iii) carrying task-relevant non-position content (capacity d_o), subject to $d_e + d_i + d_o \lesssim d$. Bottleneck encoders force $d_e \to 0$, leaving d_i to dominate the linear position code; rich encoders enlarge d_e, so policy gradient minimises d_i down to whatever residual the integration route still needs. Our Procrustes / participation-ratio / eigenspectrum-α measurements collectively probe d (effective rank ≈ 4 for blind, lower for coarse where the encoder collapse shrinks d overall, Appendix E.5); the magnitude-axis collapse follows from d_i shrinking as d_e grows. We use IB descriptively rather than mechanistically; Saxe et al. [2019]'s critique of IB-deep-learning theory cautions against absolute MI-magnitude claims, but the cross-condition *ordering* reported here (linear collapse, MLP recovery, MINE rank in Appendix E.4) is robust to those concerns.

Falsifiable predictions and broader implications. The capacity-allocation principle is architecture-agnostic: it predicts a content-level allocation in any system where a finite-capacity memory module reads a pretrained encoder. For transformer-based PointGoal navigators (e.g. Zeng et al. 2024; cf. Whittington et al. 2022 for the transformer/hippocampus correspondence), where "memory" lives in the KV cache or attention over rollout history, the principle predicts the same monotonic anti-correlation reported in Figure 2a: attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders. The cleanest falsification target is direct: train a transformer navigator under our 5 sensor conditions; observing the opposite pattern would falsify the principle's architecture-independence and bound it to recurrent-state systems. The same logic re-frames recent VLM spatial-reasoning failures [Ramakrishnan et al., 2025]: rich pretrained encoders may divert capacity from the downstream substrate's spatial code rather than merely fail to teach it, and bandwidth-restricting perception (foveation, log-polar) may be *cognition-enabling* rather than only compute-saving. The relevant axis is encoder spatial-feature *variety* per step rather than raw input bandwidth (Appendix G); a finer multi-point sweep within recurrent agents is left to future work.

Architecture-agnostic confirmation: format shift reproduces in a frozen encoder with a small LSTM stack on a different environment (Figure 12). As a partial test of the architecture-independence claim, we ported the encoder-bandwidth axis to a different environment, encoder family, and training regime: frozen DINOv2-Base [Oquab et al., 2024] patch features at five resolutions (mirroring our Habitat chassis: BLIND, COARSE, FOVEATED, UNIFORM, FOVEATED-LOGPOLAR) feed a small 2-layer LSTM trained on Memory Maze [Pasukonis et al., 2023] 9×9 offline trajectories with next-feature prediction. We then probe `agent_pos` from the LSTM hidden state (Ridge linear and 4-layer MLP, frame-level CV with 80/20 random split, eval window steps 250–500; full pre-registration in `scripts/probing/world_model_probe/PRE_REGISTRATION.md`). Three pre-registered conditions land cleanly. (1) Linear R^2 is monotone-increasing with encoder bandwidth: BLIND 0.01, COARSE 0.38, FOVEATED 0.41, FOVEATED-LOGPOLAR 0.41, UNIFORM 0.45. (2) Non-linear R^2 (4-layer MLP) is uniformly high in all sighted conditions (≈ 0.998) and at chance for BLIND (-0.0003). (3) The non-linear gap (MLP$-$linear) is monotone-*decreasing* with bandwidth: COARSE $+0.62$, FOVEATED $+0.59$, FOVEATED-LOGPOLAR $+0.59$, UNIFORM $+0.55$ — format-shift recovery is largest where linear readability is smallest, exactly the dichotomy our Habitat result reports. *Direction relative to Habitat.* The Memory-Maze setup deliberately omits the GPS/compass channel our Habitat agents receive; without that explicit positional anchor, the encoder is the only sensory route to absolute position, and BLIND cannot decode it (no signal, no integration anchor). The bandwidth-allocation *tradeoff* (where blind's integration route compensates) thus

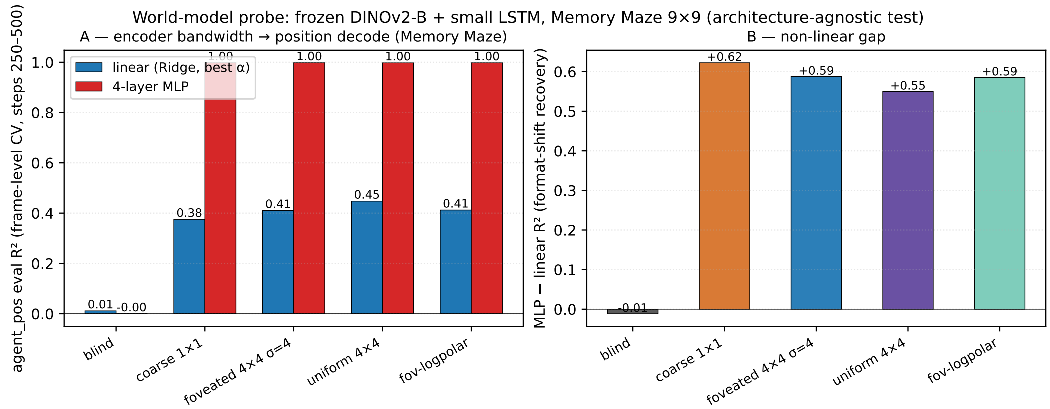


Figure 12: **Architecture-agnostic confirmation: format shift reproduces on Memory Maze with a frozen DINOv2-Base encoder feeding a small LSTM.** *(A)* Linear (Ridge, best of $\alpha \in \{0.1, 1, 10, 100, 1000\}$, blue) and 4-layer MLP (red) eval R^2 for `agent_pos` from the LSTM hidden state, frame-level random CV (80/20), eval window steps 250–500 of T=500 Memory-Maze trajectories. Linear R^2 is monotone-increasing with encoder bandwidth (BLIND $0.01 \rightarrow$ UNIFORM 0.45); MLP R^2 saturates at ≈ 0.998 for all sighted conditions and is at chance for BLIND. *(B)* The non-linear gap (MLP−linear) is monotone-*decreasing* with bandwidth: format-shift recovery is largest where linear readability is smallest (COARSE $+0.62$ vs. UNIFORM $+0.55$), the same dichotomy our Habitat agent shows in Figure 3. The opposite linear-R^2 direction relative to Habitat reflects the absence of GPS/compass channels in the Memory-Maze setup — without an alternative positional anchor, the encoder is the only route, and BLIND carries no signal.

requires a competing route; the format-shift dynamic (linear-readable code emerges from a non-linear substrate as bandwidth grows) generalises across both regimes. The two together place the broad "encoder structure shapes memory format" claim on a different environment, encoder family, and policy regime, while clarifying that the specific direction (high-bandwidth-rich vs. bottleneck-rich) is mediated by which sensory channels carry positional information.

4.4 Limitations

(i) *Single seed per condition*, following the comparative-cognition modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]. Cross-condition rigour comes from convergent multi-method evidence (linear and MLP probing, CKA, Procrustes, subspace divergence, LOSO, lag-k, 5×5 transplant, Skaggs, excursion-forgetting all rank the conditions in the same direction — unlikely under single-seed noise alone), supplemented by an independent-retraining replication of the blind condition (Appendix A). Per-condition seed-variance error bars over a multi-seed grid are reported as future work. (ii) *Method-family and content-axis scope.* Methods we do not report and that could change a finding's strength: dynamical-systems analysis (fixed-point/slow-point structure [Sussillo and Barak, 2013]; Dynamical Similarity Analysis [Ostrow et al., 2023] as a dynamics-aware Procrustes alternative) would speak to the format axis at the level of computational primitives. We piloted DSA on our h_t time-series (10-d PCA pre-reduction, 50-step trajectories) and found the cross-condition signal sensitive to HP (n_delays, rank), rollout protocol (excursion-warmup vs. deterministic-rollout), and episode-phase alignment (blind has median episode length $\sim 2.5\times$ sighted, so first-T-step samples occupy different behavioural phases) at our explored settings; the 5-condition pattern under the cleanest protocol (deterministic-rollout, all conditions, T=50) showed sighted clustering at floor ~ 0.10 angular while blind was distant (~ 0.27), but blind's within-condition split-half DSA was also ~ 0.23, leaving the SNR insufficient to publish a definitive ranking. Information-theoretic estimators of $I(h_t; \text{pos}_{t+k})$ would replace decodability R^2 with predictive-horizon nats; gaze placement and a $\sigma_{\max}$ strength sweep within the foveated condition test the rich-encoder content axis. All four are left for follow-up. (iii) *Architecture, algorithm, and task scope.* We use a recurrent (LSTM) backbone with DD-PPO, PointGoal, and a ResNet-18 visual encoder, matching the canonical pipeline of [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018] for direct comparability; the architecture choice is replication-driven rather than central to the claim, and the principle is stated at the encoder-memory interface (architecture-agnostic). A reasonable concern is whether the encoder–memory race is an LSTM artefact (e.g. vanishing-gradient gating limitations); two within-paper checks argue against it: 2-layer MLP probes recover position in rich-encoder conditions ($R^2 \in [0.51, 0.73]$), so the LSTM *does not* discard position — the linear-readable axis tracks encoder bandwidth, a property of the linear DD-PPO action head; and the cross-condition memory transplant (§3.2) is a behavioural test outside any probe framework that reproduces the same ordering. The encoder–memory race is in principle architecture-agnostic; the cleanest falsification target is a transformer-based navigator under the same conditions, identified in §4 but not verified here. The magnitude- and format-axis patterns could also shift under different RL algorithms, different navigation tasks (ObjectGoal, ImageGoal, exploration), or different visual encoders.

5 Conclusion

A five-condition visual-sensor ablation in a recurrent PointNav agent shows sensor structure as a representational-content axis whose effects cohere under a single capacity-allocation principle measured along three axes: *magnitude* (bandwidth–allocation tradeoff between encoder and recurrent-integration routes), *format* (within-condition scene-invariance, across-condition non-interchangeable subspaces with asymmetric memory transplant, and a King–Dehaene-style temporal-format dissociation in which the blind code rotates over time while the coarse code stabilises), and *consumption* (probe-readable position dissociates from policy reliance). The bandwidth–allocation ordering is preserved on held-out MP3D, blind units integrate over a $2.1{-}2.5\times$ longer Murray-cortical-hierarchy timescale than every sighted condition, and the pattern is at high level consistent with independent animal-navigation findings on hippocampal compensation under sensory deprivation, cross-modality remapping, and the no-/global-remapping axis under hidden-state inference [Sanders et al., 2020] — framed as an open bio–AI hypothesis rather than a demonstrated parallel.

Methodologically, we view the paper as a step toward a *comparative cognitive neuroscience of artificial navigation agents*: a research programme in which matched-condition populations of trained agents are analysed with the same set of metrics — population geometry, intrinsic timescales, dynamical fixed points, predictive-coding residuals, persistent-homology of the state manifold, mixed-selectivity / splitter analyses, communication subspaces, transfer entropy — that is used to map biological cortex and hippocampus. We piloted this analytic suite on the present sensor axis (Figures 7, 8, 12): temporal-axis methods (TGM, Murray timescales) and topological methods (persistent homology) detect strong cross-condition signal aligned with capacity allocation, and the format-shift dynamic itself reproduces in a frozen encoder feeding a small LSTM on a different environment. Beyond this paper, the same suite can be applied to architecture-axis populations (recurrent vs. transformer world models), training-curriculum axes (sparse vs. dense reward), or biological-vs-silicon axes (LSTMs trained on egocentric video versus rodent CA1 recordings under matched stimuli). The goal is not to claim that artificial agents *are* brains, but that the analytical methods used to ask "how does this neural system represent its world?" transfer to artificial systems where the relevant variables can be held fixed — and that the cross-condition contrasts produce theoretically informative signatures (capacity allocation, format shifts, dynamical regimes) of the same kind comparative neuroscience uses to understand sensor-niche adaptation in animals. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators (the cleanest falsification target, §4). Cross-condition rigour rests on multi-method convergence and an independent-retraining blind replication (Appendix A); a multi-seed grid, finer encoder-bandwidth sweep, and gaze-placement controls are reported as future work.

Code and checkpoints. Code is available at `https://github.com/alunxu/foveated-cog-map`; the four canonical converged checkpoints plus intermediate cross-training checkpoints (used for the §3.1 substitution-dynamics figure) are hosted at `https://huggingface.co/alunxu/spatial-memory-checkpoints`.

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Justification: §5.3 discusses implications — sensor design as a tunable axis for interpretable embodied agents. Negative societal impacts are minimal: the work is fundamental research on simulated indoor navigation with no direct deployment pathway.

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A Reproducibility: hyperparameters and implementation notes

Training hyperparameters (DD-PPO, all five conditions identical) 4-GPU `torchrun`; `num_steps=256`, learning rate `lr=2.5e-4` with `lr_decay=False`; `num_environments=16/worker`; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from Wijmans et al. [2020].

Architectural details (sensor encoding) Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (`nn.Embedding` with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the Wijmans et al. [2020] Wijmans-default sensor stack.

Implementation notes (foveated condition) The foveated condition disables the running-mean-and-variance input normaliser (`RunningMeanAndVar`) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected; the ablation is in the codebase under `src/habitat/foveated_normalised_policy.py` (normaliser *enabled*) for future work that wants to revisit this choice.

Implementation notes (numerical stability) Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) gradient sanitisation in `wijmans_policy._safe_before_step` replaces NaN/Inf gradients with zero before each PPO update; (ii) a forward-looking `geodesic_distance` clamp in `nan_safe_measures.py` prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories.

Log-polar foveation transform The log-polar control (Appendix G) uses `LogPolarFoveationTransform` in `src/habitat/torch_foveation.py`: resample the 128×128 avg-pooled image onto a 64×64 ($n_\rho \times n_\theta$) log-polar grid before the ResNet-18 backbone. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

B Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

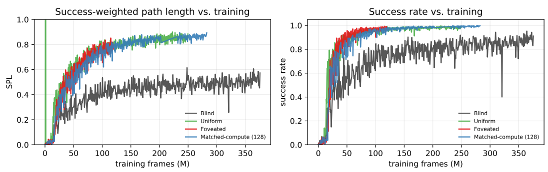
Length-matching ablation. Under deterministic rollouts, episode lengths span ∼100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the H1 ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

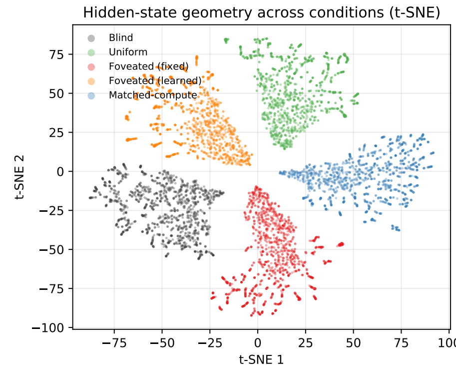
C Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states and GPS labels as the linear Ridge probe, with the same



(a) Training curves (TensorBoard event-file scalars). All three sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all four conditions (perplexity 30). The four conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§3.2).

Figure 13: Supplementary visualisations referenced from the main text.

5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, uniform 0.55 (vs. Ridge's chance R²). The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the log-polar foveation control (Appendix G) targets the encoder-spatial-output axis at one intermediate point.

D Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all four conditions ($R^2 \geq 0.81$), and the real-vs-permuted-label probe gap (Hewitt–Liang selectivity) tracks GPS R^2; the chance-level rich-encoder GPS R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix B.

Linear-probe artefacts. Two complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the 5×5 matrix's asymmetry adds structure no linear-similarity measure can read. *(b) Non-linear probe:* a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix C). H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.

E Supplementary visualisations

E.1 Capacity-allocation: additional geometric tests

The three figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-allocation (Fig. 3) and LOSO scene-invariance (Fig. 4) results in §3.1.

E.2 Single-direction β-scrubbing reveals redundant population coding

The Ridge probe identifies a 2-d direction β in $\mathbf{h}_2$ that linearly predicts (x, y). We test whether this direction is *causally responsible* for the recoverable position code by projecting $\mathbf{h}_2$ onto the null-space

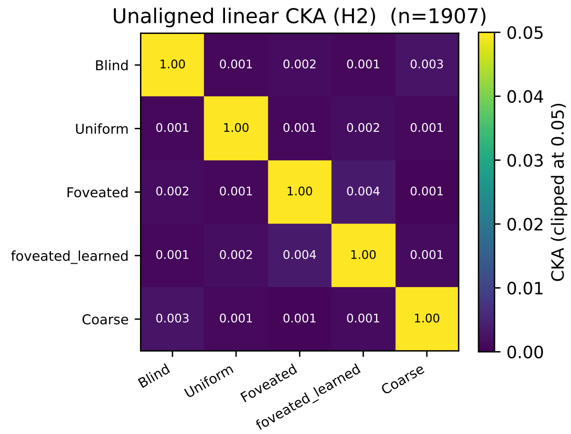


Figure 14: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 5); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

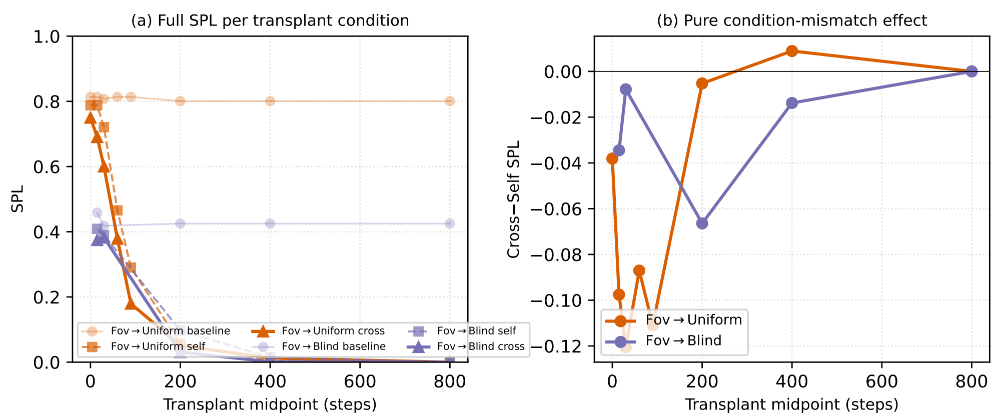


Figure 15: Memory-transplant midpoint sweep on two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps (n=100–150 episodes/cell). The full 5×5 matrix in Figure 5 (right) is at midpoint t=30. *(a):* full SPL per condition — baseline / self / cross. The t=0 self/cross collapse to a protocol-noise floor (so the diagonal in Figure 5 is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. *(b):* cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at t=30 in Figure 5 is therefore neither a t=30-specific artefact nor a late-midpoint inflation.

of β (rank-510 subspace) and re-evaluating both Ridge and MLP probes (Heimersheim & Nanda 2024 noising-style patching: tests whether β was *necessary* for the encoded behavior [Heimersheim and Nanda, 2024]).

	Linear orig	Linear scrub	MLP orig	MLP scrub
Blind	+0.95	+0.93	+0.94	+0.95
Coarse	+0.72	+0.62	+0.79	+0.84
Uniform	−1.08	−1.46	+0.42	+0.31

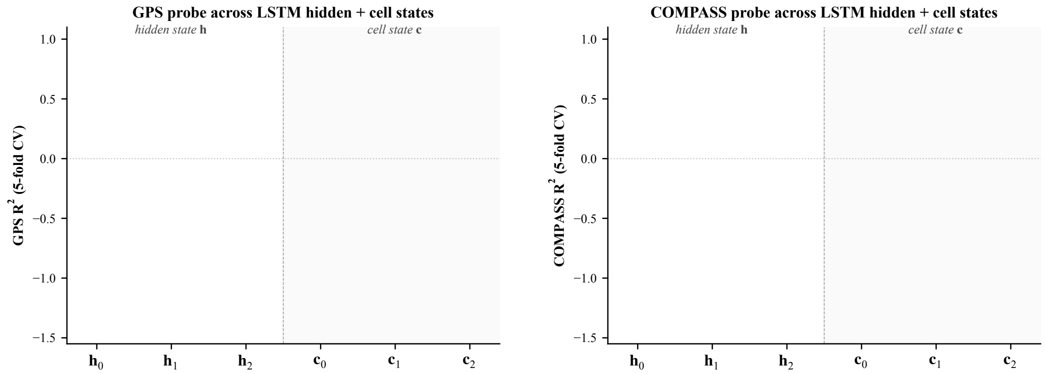


Figure 16: Linear probe (R^2, 5-fold CV) at each LSTM hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left:* GPS. *Right:* compass. Three patterns. *(i) At* $\mathbf{h}_0$ *all conditions decode GPS at* $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). *(ii)* $\mathbf{h}_1$ *dips for every condition* including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for bottleneck. *(iii) Cell states* $\mathbf{c}_\ell$ *do* not *carry the GPS code as cleanly as* $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The H1 claim about "top-layer linearly-decodable GPS" is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

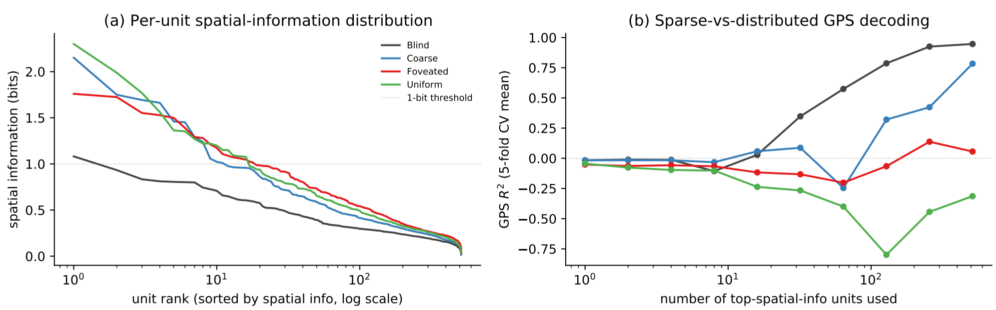


Figure 17: Population coding analysis (referenced from §**??**). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information* $\neq$ *position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

The MLP probe is essentially unaffected by scrubbing across all three conditions ($\Delta R^2 \in [-0.05, +0.11]$). The Ridge probe shows small drops for Blind (-0.02) and Coarse (-0.10), and a more-negative shift for Uniform (the absolute value increases due to the variance of negative R^2 estimates). Two interpretations: (i) the rank-2 subspace identified by Ridge is one of many redundant directions encoding position — when removed, Ridge re-fits to a different direction in the remaining 510 dims; (ii) MLP, having access to non-linear combinations of remaining dims, trivially recovers the position code from this redundant population. This is the population-coding-redundancy property familiar from biological place-cell systems [Stringer et al., 2019]: single-cell or low-rank ablation does not break the code because of high-dimensional distributed representation. Methodologically, single-direction scrubbing is too weak to causally localise position information in $\mathbf{h}_2$; higher-rank

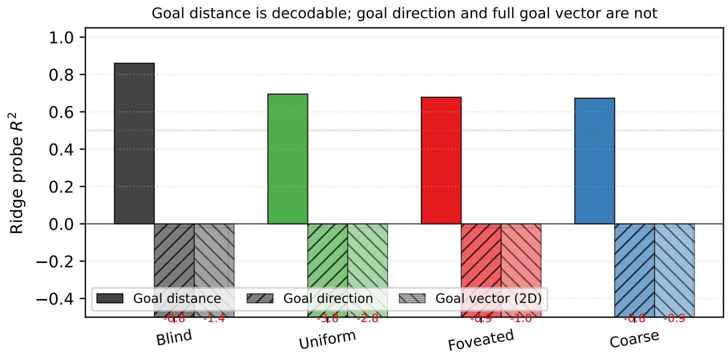


Figure 18: Goal-vector probe R^2 (referenced from §**??**). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

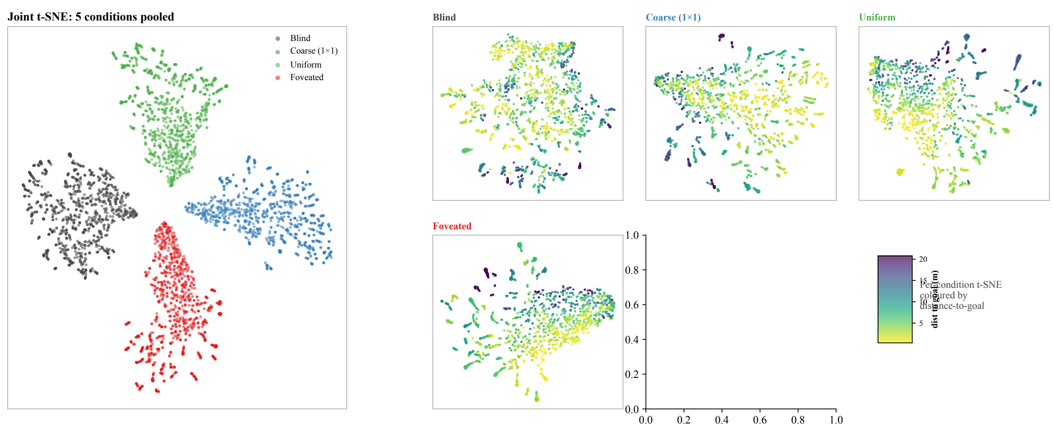


Figure 19: t-SNE of top-layer LSTM hidden states $\mathbf{h}_2$, supplementing the linear-similarity H2 results (§3.2). *Left:* joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right:* per-condition t-SNE coloured by distance-to-goal ($n = 2\,000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind's hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

ablation, attribution-based pruning, or path-patching [Heimersheim and Nanda, 2024] would be next steps. Foveated is deferred to the 250M re-probe.

E.3 Predictive horizon: does h_t encode future positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k}. We test the analog claim for our recurrent state: does a linear or MLP probe from $\mathbf{h}_t$ recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

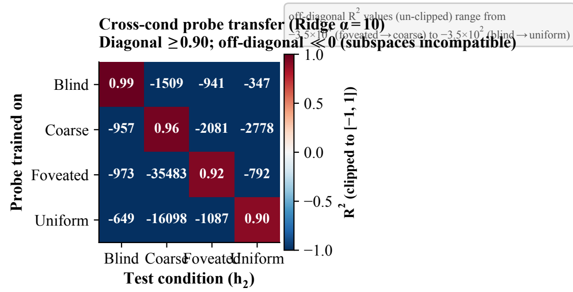


Figure 20: **Cross-condition probe-transfer matrix.** Ridge-probe trained on condition A's top-layer states, applied to condition B's. Diagonal (own-probe) $R^2 \in [0.90, 0.99]$. Off-diagonal R^2 is *catastrophically negative* ($-3.5{\times}10^2$ to $-3.5{\times}10^4$ unclipped; clipped to $[-1, 1]$ for visualisation): the position-axes encoded across conditions are not just orthogonal in subspace orientation (Figure 6) but jointly incompatible at the readout level. Even when the same scenes are visited and the same task is solved, no single linear position-direction transfers between conditions.

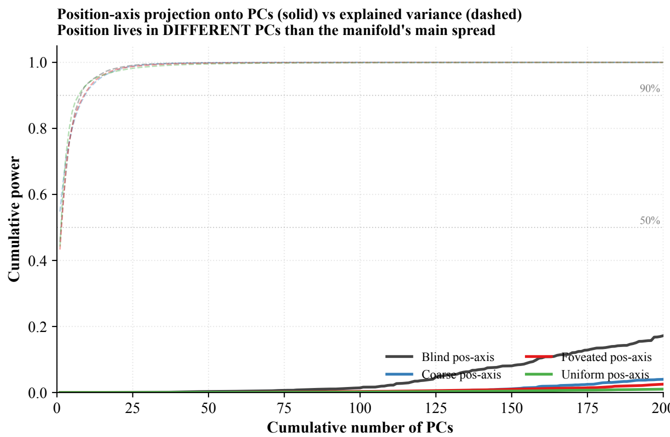


Figure 21: **Position-axis lives in low-variance directions of $\mathbf{h}_2$.** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold's own variance distribution). For all four conditions, the manifold's variance is concentrated in $\sim$ 7–9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of $\mathbf{h}_2$ across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure 4).

	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 50$
Linear (Ridge $\alpha = 10$) R^2, all-episode pooling, episode-level 5-fold CV						
Blind	$+0.94$	$+0.94$	$+0.94$	$+0.92$	$+0.90$	$+0.79$
Coarse	$+0.57$	$+0.57$	$+0.57$	$+0.56$	$+0.50$	$+0.37$
Foveated	$+0.06$	$+0.06$	$+0.08$	$+0.13$	$+0.19$	$+0.06$
Foveated-logpolar	$+0.17$	$+0.18$	$+0.17$	$+0.11$	-0.14	-0.50
Uniform	-4.31	-4.32	-4.32	-4.26	-4.39	-10.14

Three observations. *(i) Blind exhibits a long predictive horizon.* Linear R^2 from $\mathbf{h}_t$ to $(x, y)_{t+k}$ remains $\geq$ 0.90 from $k = 0$ to $k = 20$ (only 0.04-point drop), and 0.79 at $k = 50$. That is, the same linear projection that decodes *current* position from $\mathbf{h}_t$ at $R^2 = 0.94$ also decodes positions 20 steps in the future at $R^2 = 0.90$. This is a strong form of the predictive-map signature predicted by SR theory: $\mathbf{h}_t$ encodes future-state occupancy, not just current state. *(ii) Coarse follows the same pattern at lower magnitude* (linear $R^2 \in [0.50, 0.57]$ across $k =$ 0–20, dropping to $+0.37$ at $k = 50$).

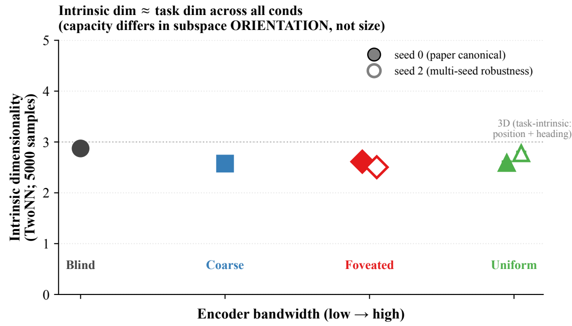


Figure 22: **Intrinsic dimensionality (TwoNN).** Facco et al. [2017]-style estimate of the intrinsic dimensionality of $\mathbf{h}_2$, on 5 000 samples per condition (paper-canonical seed-0 filled markers; seed-2 robustness check hollow). All four conditions yield ID ≈ 2.5–2.9, comparable to the task-intrinsic dimensionality (2-D position + 1-D heading $\approx$ 3). The same intrinsic dim across conditions, combined with strongly differing linear R^2 and orthogonal subspaces (Figure 6), shows capacity-allocation is about the *orientation* of the task-shaped manifold within the 512-d ambient space rather than about its size.

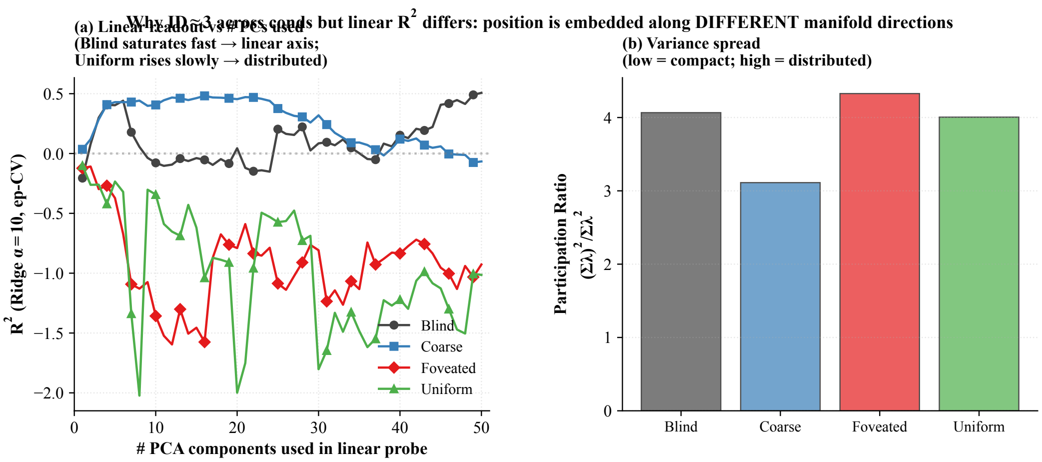


Figure 23: **Linear readout vs. # PCs.** *(a)* Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with ≤ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). *(b)* Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 21).

(iii) Rich-encoder conditions have no clear predictive horizon. Foveated stays in $[0.06, 0.19]$ across $k = 0$–50 with no monotonic decay — the small positive values reflect probe-level noise rather than a sustained future-state code. Foveated-logpolar starts at $+0.17$ but turns negative by $k = 20$. Uniform is already strongly negative at $k = 0$ ($R^2 = -4.31$) and grows more negative. Rich-encoder $\mathbf{h}_t$ encodes neither current nor future position linearly — consistent with our LOSO scene-invariance reading that rich-encoder position codes are scene-conditional and only non-linearly recoverable, but adds the temporal axis: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation.

(See Figure 32 for the predictive-horizon panel referenced from §3.1.)

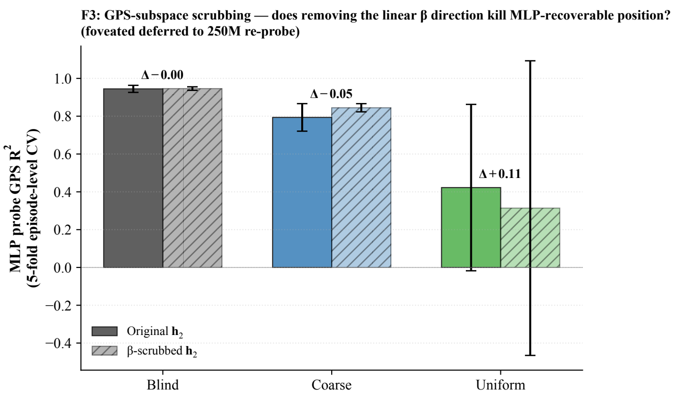


Figure 24: β**-subspace scrubbing.** For each cond (foveated deferred), Ridge β identifies a 2-d direction in $\mathbf{h}_2$ that linearly predicts (x, y); scrubbed $\mathbf{h}_2 = (I - \beta^+\beta)\,\mathbf{h}_2$ removes this 2-d subspace. MLP probe is essentially unaffected by scrubbing across all three conditions: position information is redundantly distributed across many h_2 dimensions, not concentrated in the 2-d β subspace.

E.4 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in §3.1 on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(\mathbf{h}_2; \mathrm{pos})$ per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X; Z) \geq \sup_T \mathbb{E}_{P_{XZ}}[T(x,z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x,z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired $(\mathbf{h}_2, \mathrm{pos})$ along episodes; marginal samples are obtained by within-batch shuffling of z. We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4}, 4,000 training steps per condition, and the bias-corrected gradient (eq. 12 of Belghazi et al. 2018) to mitigate the SGD bias of the log-denominator estimate.

E.5 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of Stringer et al. [2019] to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All five conditions show a tight power-law eigenspectrum (Figure 26, $R^2 \geq 0.99$). The fitted exponent shows a sharp bottleneck-vs-rich-encoder split:

Condition	α	R^2	Var. top 10 PCs
Blind	3.85 ± 0.03	0.99	95%
Coarse	1.80 ± 0.01	1.00	82%
Foveated	1.72 ± 0.01	1.00	76%
Foveated-logpolar	1.80 ± 0.01	1.00	95%
Uniform	1.90 ± 0.01	1.00	91%

Two observations. (i) Blind sits well above the critical exponent $1 + 2/d_{\text{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\text{task}} = 3$ stimulus (current position + heading), and far above Stringer's V1 value ($\alpha \approx 1.04$). With $\alpha = 3.85$, blind's eigenspectrum decays sharply: variance is concentrated in the top few PCs (top-10 already capture 95%). We do not claim that Stringer's specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d-dim stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) The four sighted conditions cluster tightly around $\alpha \approx$ 1.7–1.9, all near the $1 + 2/d_{\text{task}} \approx 1.67$ critical exponent — the smoothness/efficiency border. Blind is the outlier above, in a few-dim concentrated regime, while the four sighted conditions distribute variance across more PCs. The split

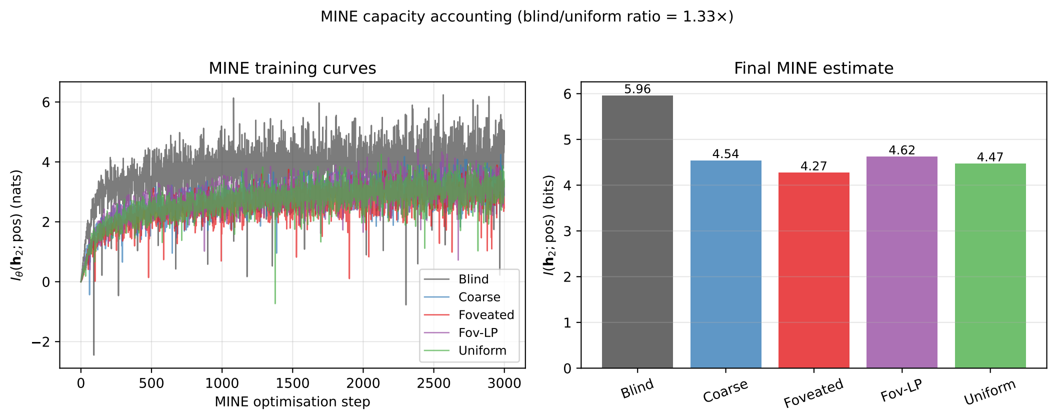


Figure 25: **MINE capacity accounting.** *(a)* MINE training curves: $I_\theta(\mathbf{h}_2; \text{pos})$ in nats vs. optimisation step, per condition. All curves converge by $\sim 3{,}000$ steps. *(b)* Final $I(\mathbf{h}_2; \text{pos})$ in bits per condition (mean $\pm$ std across 3 MINE seeds): Blind 6.01 ± 0.06, Coarse 4.55 ± 0.03, Foveated 4.30 ± 0.04, Foveated-logpolar 4.66 ± 0.04, Uniform 4.47 ± 0.06. Total mutual information decreases with encoder bandwidth, but the decrease is gentle ($\approx 1.3\times$ blind/uniform) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The within-sighted ordering (foveated $<$ uniform $<$ coarse $<$ foveated-logpolar) is reproducible across MINE seeds (per-cond std ≤ 0.06 bits, much smaller than the within-sighted gap ≈ 0.36); the dominant signal is the bottleneck-vs-rich-encoder gap (Blind 6.01 vs sighted mean 4.49). The interpretation: rich-encoder conditions retain substantial position information in $\mathbf{h}_2$, but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure 4).

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-4, unified hp)
Encoder feature-map probe	where position information is held	**landed; see §4**

bottleneck-vs-rich-encoder reading: blind's heavy reliance on the integrated GPS code crystallises into a low-dimensional eigenspectrum, while sighted conditions (including coarse, whose 1×1 encoder still feeds a ResNet-18 stack with diverse channels) distribute hidden-state variance across more directions to support encoder-route information.

E.6 Persistent-memory failure-mode catalog

The main-text Figure 35 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 27) shows four cases per condition spanning the per-condition margin range (most-tries-new $\to$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5\,\text{m}$ (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

F Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from the main paper. All four are submitted.

Foveation strength sweep (F1–F4) — planned for follow-up, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at

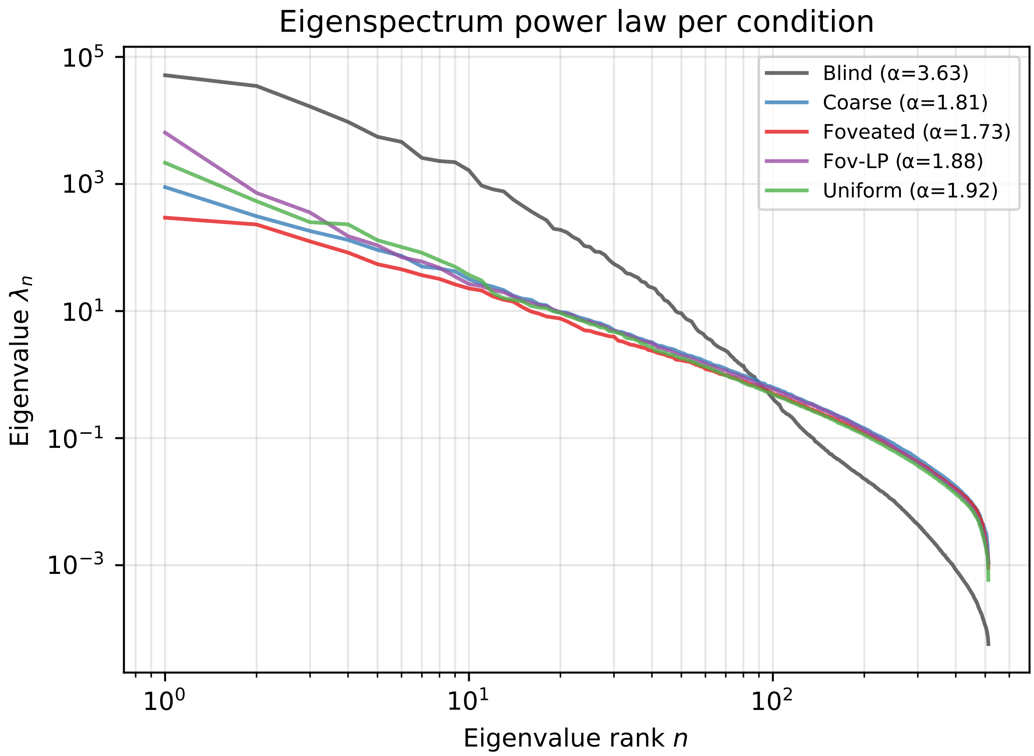


Figure 26: **Eigenspectrum power-law per condition.** *(a)* Eigenvalue λ_n vs. PC index n for $\mathbf{h}_2$, log–log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. *(b)* Fitted power-law exponent α per condition (linear fit on log-λ_n vs. log-n, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.

low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 4×4 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2\times2$, much closer to coarse's 1×1 than to uniform's 4×4. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. The preregistered prediction (written before result was available) was that log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.58$ and uniform's ≈ 0, with corresponding behavioural shifts; if log-polar matched uniform, the encoder-spatial-output-dimensionality mechanism would be wrong. The result (next paragraph) falsifies that simple reading.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper's interpretation.

G Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The

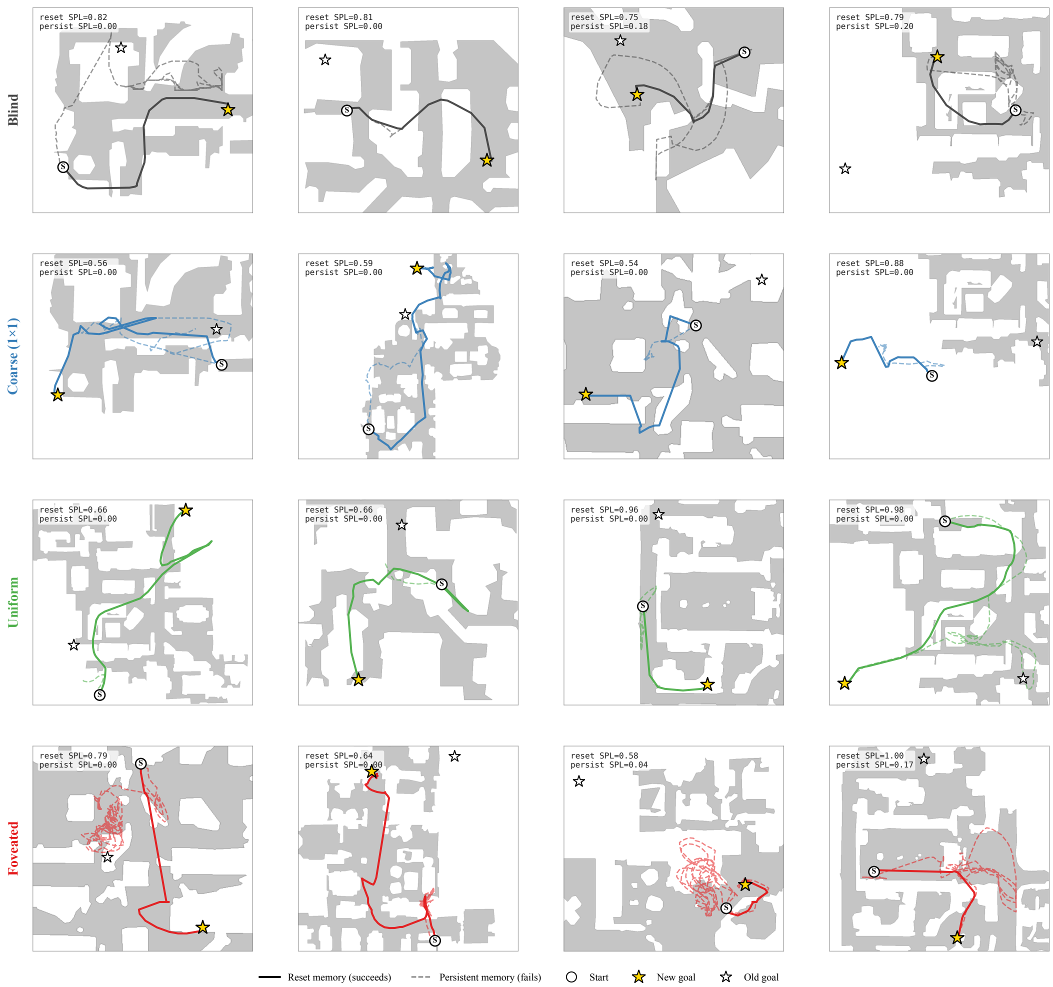
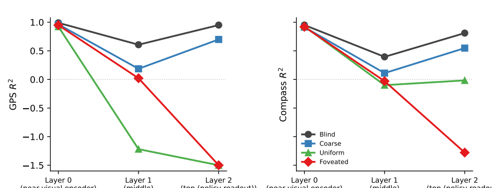
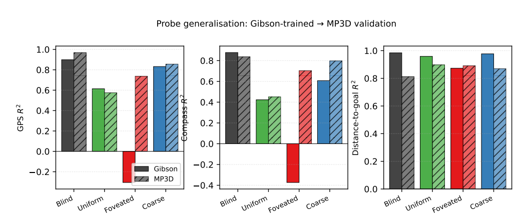


Figure 27: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at "tries-new" (most cases attempt new goal direction but fail to reach); Uniform clusters at "locks-onto-old" (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single seed per condition (cogsci modelling convention; see §4.4).



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all four conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 30a (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-rich-encoder partition is the same.

Figure 28: Per-layer and MP3D companion figures (referenced from §3.1).

most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2 \times 2$, intermediate between coarse's 1×1 and

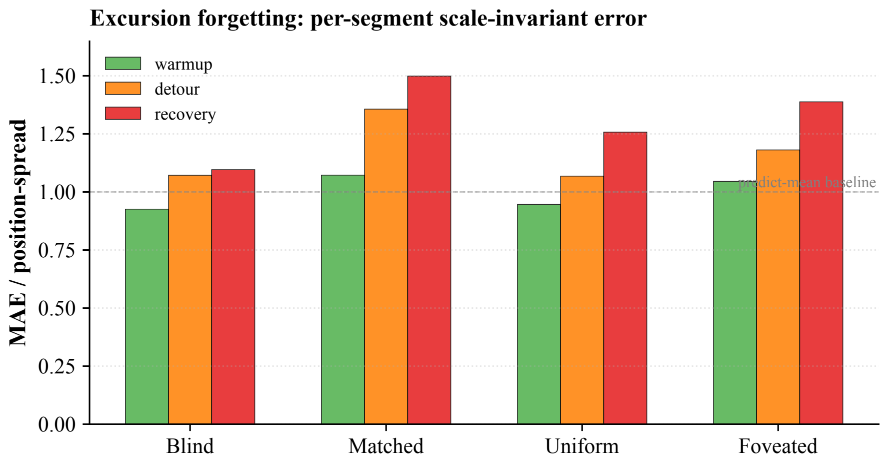


Figure 29: Excursion-forgetting per-segment scale-invariant error (referenced from §**??**). One Ridge probe trained on full-episode pooled hidden states (episode-level 5-fold CV); the same probe tested on each segment of the held-out fold. Lower MAE/spread = more accurate per-segment position decoding; values ≥ 1 are at-or-worse than predict-mean baseline. All five conditions show recovery $>$ warmup error (forgetting); blind's rise ($+0.381$) is the largest, roughly 1.8–$2.8\times$ the sighted conditions ($+0.135$ foveated-logpolar, $+0.167$ uniform, $+0.198$ coarse, $+0.210$ foveated). The pattern is consistent with blind carrying the most action-stream-dependent integrated GPS code and the sighted conditions partly re-anchoring on visual landmarks during recovery. Single seed.

uniform's 4×4 (verified by direct forward pass through the trained encoder). Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered prediction:** if the H1 mechanism is the simple "encoder spatial-output dimensionality" reading, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse's $+0.58$ than to uniform's ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per step, or another axis. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-49, $\sim$ 245M frames) yields top-layer GPS $R^2 = -0.029 \pm 0.759$ (5-fold CV, $n = 500$ episodes) — *not* in the bottleneck regime, but indistinguishable from foveated ($R^2 = +0.178 \pm 0.659$) and clearly far from coarse ($R^2 = +0.582 \pm 0.099$). The simple "encoder spatial-output dimensionality" framing is therefore *not* sufficient to explain the H1 phenomenon: log-polar's 2×2 output does not place it in the bottleneck regime. We read the result as supporting the alternative "encoder spatial-feature *variety* per step" framing: log-polar's radial sampling preserves rich gaze-centred feature variety per timestep even at low spatial output, so the policy can still source position from the visual route and the LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure 2b) corroborates this reading: log-polar starts at $R^2 = +0.92$ at $\sim$ 50M frames (matching all 4 main conditions' early high R^2), then decays along the rich-encoder trajectory rather than plateauing in the bottleneck range. A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at $\sim$ 250M frames each) was outside this submission's compute window.

H Additional representational and behavioural diagnostics

This appendix collects companion figures that extend results reported numerically in the main text. They are not load-bearing for any main-text claim and may be skipped on first reading.

H.1 Magnitude-axis: panel-level detail and predictive-horizon panel

The §3.1 consolidated 3-panel figure (Figure 2) summarises the magnitude-axis story; the figures below preserve the panel-level detail referenced from that section.

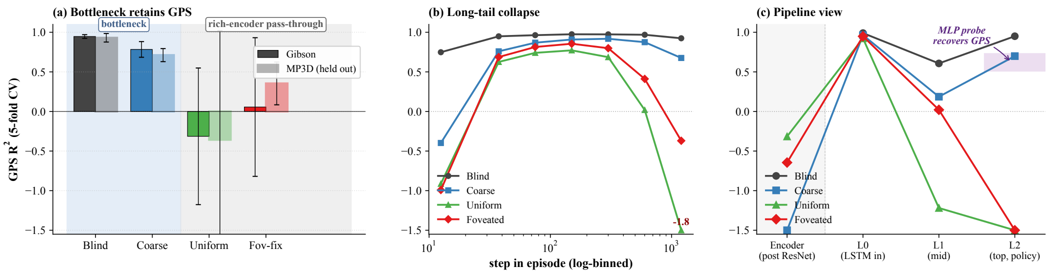


Figure 30: **Magnitude-axis evidence in detail (referenced from §3.1 and Table 1).** *(a)* Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; values outside y-range labelled). *(b)* Top-layer GPS R^2 binned by step-in-episode (7 log-spaced bins in $[10, 1600+]$, right-most bin pools ≥ 1024); rich-encoder conditions hold $R^2 \in [0.6, 0.8]$ in the typical mid-episode window then collapse in the long tail. *(c)* GPS R^2 along Encoder $\rightarrow$ L0 $\rightarrow$ L1 $\rightarrow$ L2 (this panel is condensed into Figure 2c; the version here uses single train/test fits rather than 5-fold CV mean). Shaded band at L2: 2-layer MLP probe recovery range ($R^2 \in [0.51, 0.73]$).

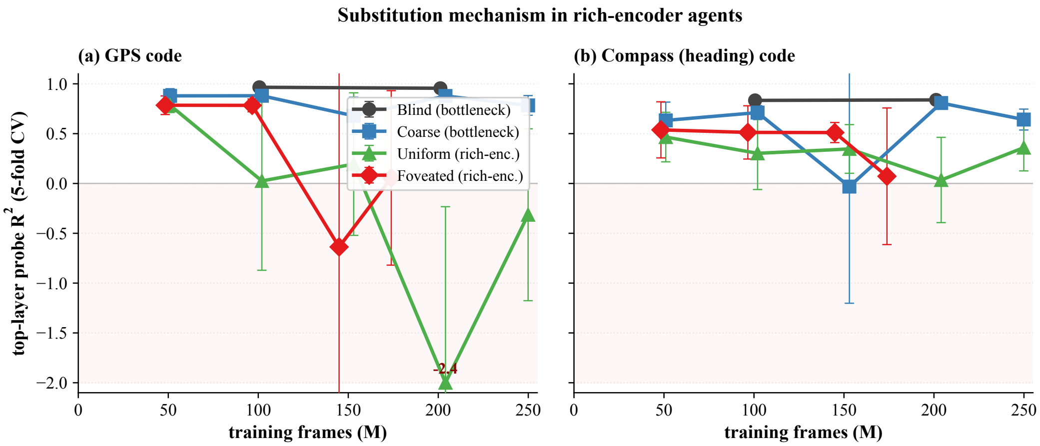


Figure 31: **Cross-training substitution dynamics: GPS and compass (referenced from §3.1).** Probe R^2 across 4–5 training checkpoints per condition, truncated at 250M frames. *(a)* Top-layer GPS R^2 (this panel is the source for Figure 2b). *(b)* Top-layer compass R^2 tells the same story at smaller magnitude — the substitution applies to spatial codes broadly, not just position. (500 ep/probe, 5-fold CV, single seed; hollow dotted = partial-data ckpts; negative-R^2 outliers clipped at -2.0.)

H.2 Other diagnostics

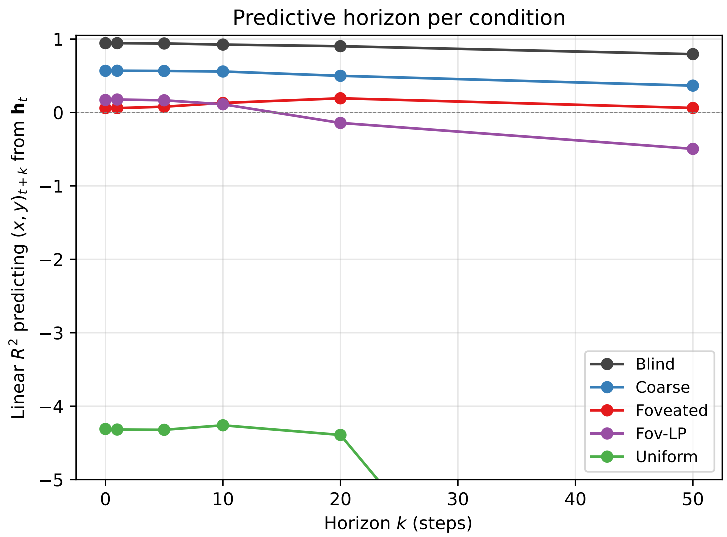


Figure 32: **SR-style predictive horizon (referenced from §3.1 and §E.3).** Linear R^2 predicting future position $(x, y)_{t+k}$ from $\mathbf{h}_t$ vs. horizon k (Stachenfeld 2017 SR-style cognitive-map signature). BLIND retains $R^2 \geq 0.90$ to $k{=}20$ and $+0.79$ at $k{=}50$; COARSE a moderate $R^2 \in [0.50, 0.57]$ to $k{=}20$ ($+0.37$ at $k{=}50$); rich-encoder conditions show no predictive horizon (foveated stays in $[0.06, 0.19]$ flat, foveated-logpolar turns negative by $k{=}20$, uniform is strongly negative throughout). Blind's $\mathbf{h}_t$ encodes future-state occupancy — the canonical cognitive-map signature predicted by SR theory.

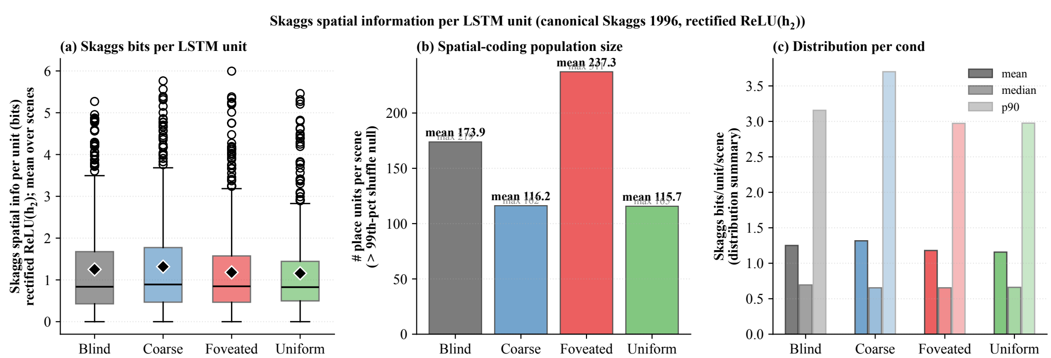


Figure 33: **Skaggs spatial information per LSTM unit (referenced from §3.1).** *(a)* Distribution of per-unit Skaggs bits (averaged across 15 top scenes per condition); bottleneck and rich-encoder conditions overlap substantially (means $\approx$ 1.16–1.32 bits). *(b)* Number of place units per scene exceeding the 99th percentile of a 100-shuffle null; foveated has the largest count ($\approx$ 237/scene), blind/coarse/uniform $\approx$ 116–174. *(c)* Distribution summary; cross-condition differences sit within the within-condition spread. The canonical Skaggs metric on rectified activations does not by itself discriminate bottleneck from rich-encoder regimes; the per-unit-regression analysis in main-text §3.1 (trajectory-phase vs. raw-position R^2) is the more diagnostic reading.

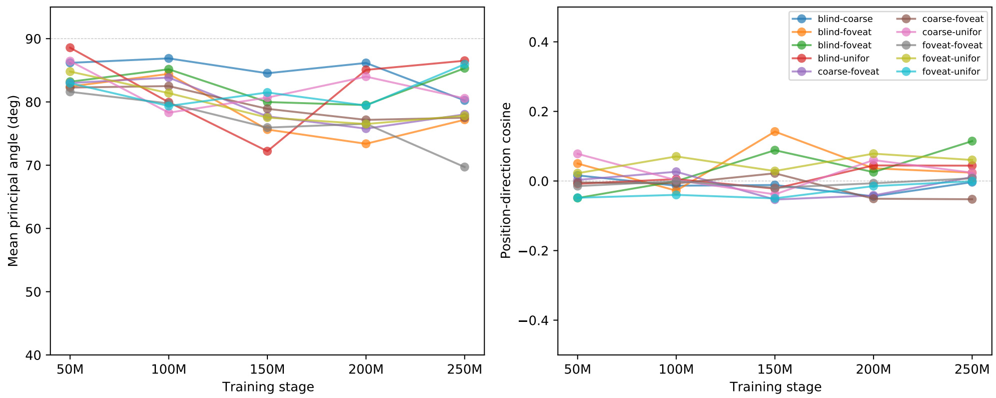


Figure 34: **Subspace evolution over training (referenced from §3.2).** *(a)* Mean principal angle between condition-pair top-K subspaces, plotted across five training stages ($\sim$ 50M to $\sim$ 250M frames; sighted at ckpt 10/20/30/40/49, blind at 10/15/20/25). Angles are already $[82, 89]$ at the earliest probed stage and stay near-orthogonal throughout. *(b)* Position-encoding direction cosines are essentially zero throughout (range $[-0.05, +0.14]$), indicating direction divergence is committed before the earliest probed checkpoint. Single seed; 4–5 ckpts per condition. Companion to main-text Figure 6.

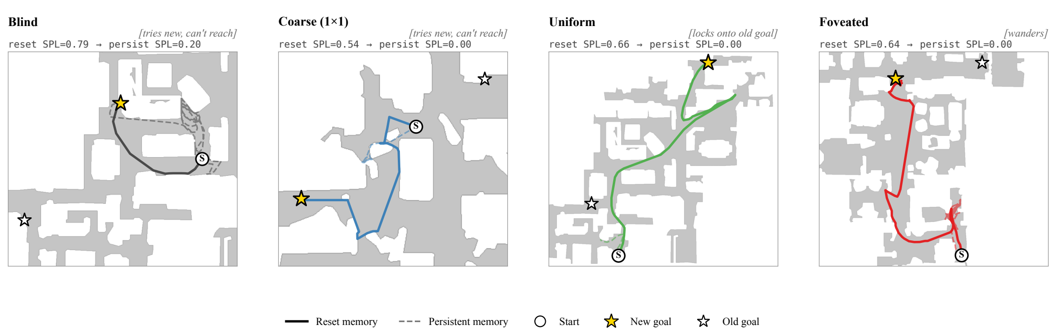


Figure 35: **Persistent-memory failure modes (referenced from §3.4): canonical paired-episode case per condition.** Extended catalog: Figure 27. Aggregate per-condition margin = avg dist(persistent end $\rightarrow$ old goal) $-$ avg dist(persistent end $\rightarrow$ new goal) over same-floor failures (reset SPL $>$ 0.5, persistent SPL $<$ 0.2, persistent steps $\geq$ 30, $|\Delta y_{\text{goal}}|$ $<$ 0.5m): Blind -0.38m (n=27), Coarse -0.57m (n=35), Foveated -0.59m (n=16), **Uniform** $+1.83$**m (**n=46**, closer to OLD goal)**. Single seed.